



An optimized intrusion detection system for resource-constrained IoMT environments: enhancing security through efficient feature selection and classification

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Abstract

This paper presents a resource-efficient IDS tailored for the IoMT, primarily emphasizing IoMT-enhanced security in medical networks. Characteristically resource-constrained and inherently vulnerable to various cyber threats, IoMT devices are typically ordinary. Inherent IoMT IDS pays off with the necessity for a critical trade-off between detection accuracy and computational efficiency. For this reason, the two-step feature selection stage is proposed to bring this challenge down. Firstly, the system decreases the dimensionality by operating the MI filtering on the dataset and keeps only the most informative features. It further refines this using ensemble-based ranking methods, such as Random Forest, AdaBoost, XGBoost, and LightGBM, to ensure the optimum feature selection for the task. Random Forest was adopted for the final classification because it is generally robust, efficient at handling high-dimensional data, and usually performs well. The proposed system has been tested with intensive usage using three well-acknowledged benchmark datasets, namely WUSTL-EHMS-2020, NSL-KDD, and CIC-IoMT2024. It showed considerable accuracy, precision, recall, and F1-score gains, particularly for DDoS and DoS attack types. The proposed technique significantly reduces execution time and memory usage. This, in turn, makes the approach much more suitable for real-time implementation. At the same time, the gains to be had in computational resource savings make this an attractive approach for resource-constrained hardware environments. This work proposes an effective and efficient IDS for IoMT security.

Keywords Intrusion detection system (IDS) · Internet of Medical Things (IoMT) · Feature selection · Resource-constrained IoMT

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1 Introduction

IoMT changed the face of the health sector by enabling different medical devices, sensors, and healthcare systems to connect to the Internet to gather information about patients' conditions for diagnostic and treatment purposes with speed [1]. During the challenges, the security of IoMT systems has become an important concern because of increased availability and deployment in IoMT critical healthcare environments. Given the critical nature of healthcare data, an IoMT device breach can lead to life-threatening consequences. The security of such networks, therefore, is extremely crucial. IDS has thus been a primary tool in IoMT systems' defense against all kinds of attacks by detecting, in real-time, unauthorized accesses, anomalous behaviors, or potentially harmful actions. Nevertheless, for it to be effective, its capability in a similar environment should deal efficiently with voluminous data; the resource-constrained nature of the IoMT devices exacerbates this situation. Therefore, feature reduction techniques have emerged as a key driver for better performance of IDS, with reduced computational load and without sacrificing high detection accuracy. The following paper will introduce a new method for reducing features [2]. This was tested on benchmark datasets and outperformed any other existing method regarding memory, CPU efficiency, and RAM. IoMT networks assemble wired/wireless interconnected medical devices and sensors to collect, transmit, and analyze patient health status data. Wearable sensors, implanted medical devices, and stationary systems such as diagnostic and hospital information systems are all part of IoMT networks. However, while IoMT has many advantages, these networks are predisposed to several security risks, from data breaches to device tampering to DoS attacks. Where attackers target the revelation of sensitive patient data and also for sabotage using medical devices, robust security concerns have evolved into a great priority for healthcare providers and patients [3–5]. Intrusion Detection Systems, or IDS, are designed to monitor and analyze network traffic and system behaviors to detect malicious activity or policy violations. In addition, the amount of data from IoMT devices tends to increase significantly when interconnected devices generate enormously high volumes of data, thereby increasing the requirement for real-time monitoring. Conventional IoT subjects IDS to massive challenges, especially in the case of IoMT. This goes much more with the limited computational resources on IoMT devices that challenge conventional IDS. The features used in intrusion detection are expected to be relevant, implicitly suggesting the applicability of quality; IDS performance generally depends on the system's efficient processing of such features [6]. Feature reduction methods become inevitable for IoMT environments. In this context, their role is to increase the performance of IDS through dimensionality data reduction, reducing memory usage, CPU load, and energy consumption [7]. Feature reduction is the process or technique for selecting a subset from an original dataset as the most relevant features, by which the goal may be pursued is that the performance of machine learning algorithms will be improved and computational resource requirements remain small. This is especially relevant in

the context of IDS for IoMT, given that these devices' bounded processing power, memory, and energy necessitate efficient data handling. It has the potential to confer greater efficiency on IDS deployed on resource-constrained devices with no compromise in detection accuracy by shrinking the number of features [8]. Various feature reduction methods, such as PCA and wrapper and filter techniques, have been proposed in the literature [9, 10]. Therefore, deploying traditional IDS [11] would become difficult as these are usually designed to operate on systems with plenty of computational resources [12]. IoMT devices operate in real time; hence, the detection and response against security threats severely affect patient safety [13]. Feature reduction is considered one of the most promising approaches to improving efficiency for IDSs in IoMT environments since the data to be processed by the IDS are reduced directly. Thus, memory, CPU, and energy consumption are reduced directly. An investigation of the trade-offs between computational efficiency and detection accuracy is needed to gain a more profound insight into the importance of feature reduction in intrusion detection systems for Internet of Medical Things environments [14, 15]. As mentioned, IoMT devices face very tight resource limitations; intrusion detection performance directly depends on how well it can handle such limitations. The most discriminating goal of a feature reduction technique is to seek an optimal trade-off between computational efficiency and accuracy in the detection model. By removing a few irrelevant or redundant features, IDS could save their efforts on the most informative part of the data and improve the performance of the whole computing process [16]. Still, intrinsic risks come with removing too many or erroneous features, which may degrade the ability to identify and classify attacks accurately, including novel and sophisticated attacks [16–18]. A standard approach in machine learning-based IDS is to avail algorithms for feature selection that rank as important features according to their contribution to their model's performance. Feature ranking techniques, such as RFE, mutual information, and random forests, have been commonly used. These methods contribute not only to a reduction in size concerning the input data but also to creating more interpretable models from the machine learning point of view, which is an important feature in healthcare environments. Feature reduction forms an essential part of the efficient intrusion detection system using IoMT and, therefore, can be said to help reduce the dimensionality of high-dimensional data that earlier used to remain unmanaged because of their uncertainty and complexity. This helps form efficient IDS that can run on resource-constrained devices without compromising detection accuracy. Therefore, different feature reduction methods must be developed to target the IoMT environment so that the medical devices remain secure and the patients are safe. Ultimately, feature reduction shall be an imperative part of scaling IDS within expanded IoMT networks to adapt to new threats accordingly [19]. Figure 1 depicts three critical aspects of IoMT.

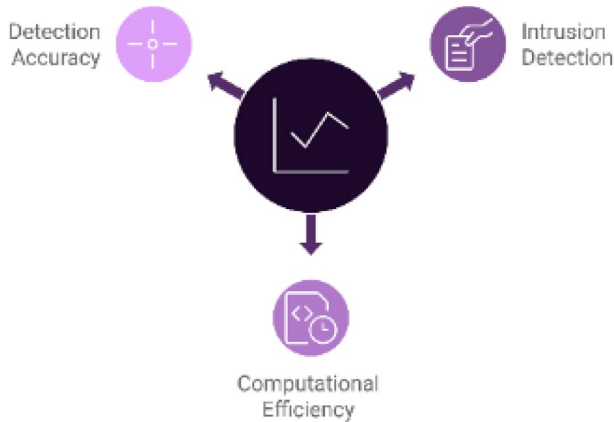


Fig. 1 Core components of IoMT

1.1 Main contributions of this paper

This paper, therefore, deals with one of the most critical needs of cybersecurity in IoMT by addressing two significant contributions: proposing a resource-efficient yet reasonably accurate IDS and selecting novel ways of feature selection and classification that optimize computational efficiency with detection performance in unique IoMT environments characterized by resource-constrained devices and sensitive network traffic.

1.2 Two-step feature selection process

One of the valuable contributions of this paper is the development of a very efficient feature selection methodology in a highly resource-constrained IoMT environment. The method is based on two steps: selecting features with a preliminary reduction of dimensionality from MI filtering and further ranking top features using an ensemble-based set of random forests, AdaBoost, XGBoost, and LightGBM. Combining these two techniques will balance the system and reduce the data's dimensionality without losing much accuracy. Hence, this two-tier feature selection technique directly addresses the computational limitations in IoMT devices, meeting the trade-off between complexity and performance. This new ensemble-based ranking applied after filtering using MI will ensure that, from all features, only those relevant to classification will be kept, hence reducing overheads emanating from medical devices.

Optimized classification with random forest A Random Forest classifier was used in the paper for the final classification stage. The basis of the classifier is Random Forest, which is considered a solid selection due to its inherent robustness to overfitting, allowing high-dimensional data, and appropriateness for complex decision boundaries that are very common in cybersecurity. Using Random Forest ensures an IDS can provide high accuracy at a small computational footprint. This framework was then

tested and further analyzed by using three benchmark datasets, namely WUSTL-EHMS-2020, NSL-KDD, and CIC-IoMT2024, which checks the correctness, recall, precision, and F1-score values for the most substantial attack-type detection.

High performance and low computational overhead This work also contributes by keeping a high-performance level, drastically reducing the amount of memory used and CPU demand. Hence, such a system holds great potential to be deployed in real-time IoMT environments. The feature reduction techniques used herein differ from those used and demonstrate that the system operates within the average resource-constrained capacity of medical devices without compromising detection performance. Specifically, this is very important for healthcare-oriented applications, as real-time threat detection—the safety of subjects and operational efficiency—needs special attention. This paper further confirms that the proposed IDS maintains detection accuracy above 95% but consumes resources much less than the traditional models.

Extensive evaluation of multiple datasets Another parameter that further seals the contribution of this paper is the rigorous evaluation it conducts using multiple datasets, such as CIC-IoMT2024, WUSTL-EHMS-2020, and NSL-KDD. Any empirical evidence for robustness and adaptability in the system would thus be due to the various tests conducted with different datasets characteristic of real IoMT traffic and simulated attacks. Whereas it performs well in general attack detection, including DDoS and DoS, its performance also rises regarding the other less frequent types of attacks, such as Remote-to-Local and User-to-Root, which were wholly trickier to detect by using traditional IDS models. Large-scale testing will ensure the overall scalability and versatility of the proposed IDS for different IoMT applications and under various threat environments.

Implication for real-time deployment in IoMT One of the critical contributions that this paper provides is practical feasibility for a real-world IoMT environment. The most significant challenge in the deployment of IDS on resource-constrained medical devices is addressed by significantly reducing the memory and computational needs of the IDS without compromising detection accuracy. It thus forms a highly suitable solution for real-time IoMT deployments, where any delay in detection and response against cyber threats may have life-threatening consequences. This contribution shows the proposed IDS's capability to protect medical networks without causing resource over-processing for devices with limited resources. The contribution represented by this paper is excellent in terms of the IoMT intrusion detection system since it is a solution that balances computational efficiency with detection accuracy. Therefore, the novelty of the feature selection method optimized Random Forest classification and proper validation on multiple datasets constitutes some of the main contributions pushing cybersecurity boundaries for all resource-constrained devices.

2 Related works

Elmasry et al. [20] propose a new method for improving intrusion detection systems of networks. The study addresses challenges brought about by the redundancy of dataset features, such as a high false alarm rate and low detection accuracy. Besides,

it suggests a dual PSO algorithm in selecting feature subsets optimized with their hyperparameters. The experiment applied the proposed approach to deep learning models: Deep Neural Networks, Long Short-Term Memory, and Deep Belief Networks. The validity of the proposed approach with drastic improvements in detection rates was performed with about a 4–6% increase and a reduction in the false alarm rate by about 1–5%, using two datasets: NSL-KDD and CICIDS2017. They also carried out an extensive collection of experiments: binary and multiclass classifications, using double PSO, which remarkably outperforms the models compared to when no pretraining is used. This provides an excellent methodology to enhance the accuracy and efficiency of NIDS in identifying complex cyber threats.

In [19], the authors have proposed a hybrid methodology for enhancing intrusion detection system performance within the IoMT environment. The authors address the challenges of security vulnerabilities attributed to various cyber-attacks, Document Object Models, probes, and phishing attacks, which might compromise sensitive medical information. They further emphasize that this model guarantees secure IoMT architecture data transfer, a high-end criticality issue preventing all possible life-threatening incidents due to potential data breaches. This contribution thus provides a concrete base for the fast-expanding IoMT security area by integrating optimized feature extraction methodologies with deep learning models. It characterizes enlarged scalability and real-time applicability for medical cloud data protection.

Paper [21] proposes an ML-based enhancement of data processing for IoMT environments. The real-time management of stress levels is focused on fog computing, not overloading cloud systems with data. It presents a two-phase ML model that first detects incoming data priority and performs causes-of-stress classification in the second phase, such as interruptions or time pressure. The fog layer processes high-priority data. The model optimizes resources, lowering latency, which is crucial for real-time healthcare applications. Out of four ML models tested, including KNN, SVM, Bagged Tree, and ANN, the latter showed better results regarding the speed of prediction and accuracy, reaching a 99.97% F1-score. The proposed framework can also be applied within the health sector with possible Vizcom scalability and efficiency in the IoMT environment, where decision making is based on real-time data.

Zhenyang Sun et al. [22] describe, in the feature selection process, 12 relevant features are selected to improve detection accuracy. Besides, it enhances the precision and recall. Among them, AdaBoost outperforms others with a recall value of 0.96. The proposed hybrid PSO-AdaBoost leading system can be considered part of machine learning-based approaches for IoMT environments, enhancing device communications and patient data security. Henceforth, it carries practical utility in building robust security protocols in IoMT systems concerning reliable health monitoring and the safety of patient information. Using machine learning algorithms such as AdaBoost underlines new opportunities that can be explored to develop more effective solutions to intrusion detection that respond to the unique challenges arising from interconnection in healthcare systems.

Han et al. [23] describe an essential contribution to the energy efficiency and sustainability of IoMT systems. The authors propose a clustering model for

medical applications, named CMMA, specifically designed to present the central issue of energy consumption in medical devices in the case of remote health care. Indeed, the proposed CMMA refines the CH selection process based on the five parameters: energy, delay, distance, capacity, and queue, and enhances the lifetime and sustainability of edge-computing IoMT systems. The result of an extensive simulation shows that the proposed CMMA outperforms earlier clustering algorithms, namely LEACH and PSO, due to energy efficiency and even distribution of cluster heads, which prolongs the lifetime of a network for practical operation. The number of cluster heads is dynamically adjusted by the proposed method concerning the transmission range, hence suitable in both small- and large-scale networks. This contribution finds its application in healthcare applications that demand reliable energy-efficient data transmission. This work has become pivotal in IoMT systems' performance enhancement; therefore, any future works involving energy-efficient communication in IoT environments for healthcare would be complete by referring to the work presented here.

Namrata Singh and Ayan Kumar Das [24] present a scalable, energy-efficient approach toward data offloading in the Internet of Medical Things systems. The authors have proposed a four-tier cloud-fog architecture to resolve energy consumption, latency, and resource utilization challenges in IoMT environments. The architecture used in this work adopts fuzzy logic for data classification based on the severity of the medical data, like emergency, significant, and general; MQTT protocols to ensure secure and ordered data dissemination; and an adaptive scheduling method to ensure effective load balancing among fog nodes. To this end, clustering of the fog nodes is performed based on their computing capacity and residual energy to optimize the particular distribution and reduce latency. The proposed approach in this work shows that the simulation using the iFogSim toolkit reduces energy consumption by 77% while significantly improving end-to-end latency and resource utilization over the existing approaches. This work will suit real-time health monitoring system applications requiring high reliability, low latency, and energy efficiency. It provides an overall framework for the enhancement of performance in IoMT systems. Hence, it can be a crucial reference in improving IoT-based healthcare solutions.

The paper [25] presents a system contributing to the innovation of two of the most critical challenges in e-healthcare: energy consumption and biosensor network data security. Based on IoMT, this framework focuses on how a biosensor node optimizes communication overheads and minimizes energy usage securely during healthcare data transmission. This architecture embeds WBAN and Kruskal's algorithm techniques to optimize decisions on routing for higher energy efficiency and lower data transmission cost. The proposed solution leverages lightweight cryptographic methods that enhance the security of patient data and ensure integrity and privacy against possible threats. The simulation shows significant performance improvements: Energy consumption is reduced by 29%, and the link breaches are decreased by 48% compared to the other existing frameworks. The framework also results in an 18% improvement in network throughput and a 44% decline in the packet loss rate. This work thus significantly addresses two significant facets of the energy efficiency and data

security requirement toward reliability and effectiveness in IoMT-based digital healthcare applications.

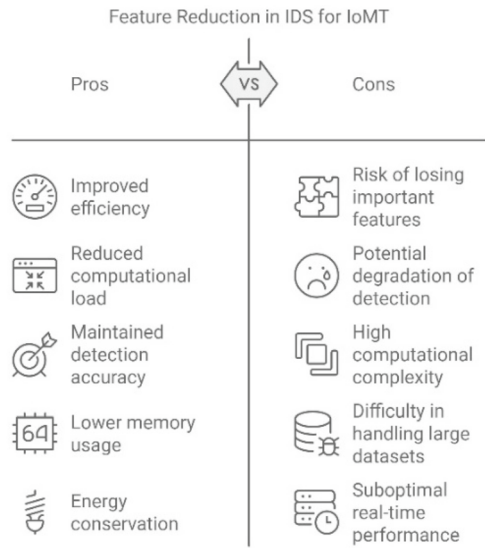
The contribution of this paper [26] is centered around reducing the energy spent in a biosensor node without impacting data integrity with lightweight crypto techniques. This significantly reduced the demand for energy consumption through Kruskal's algorithm to optimize routing. The latter architectures underwent various tests and resulted in an improvement in energy consumption by 29% and a reduction in security breaches by 48%. They also outperformed the network performances, increasing throughput by 18% and reducing the system's packet loss rate by 44%.

The work conducted by [27] presents that the proposed hybrid approach combines knowledge from network traffic data with patient biometric data from the WUSTL-EHMS dataset for anomaly detection without relying on known attack signatures. The best performance was obtained using the RFE-Decision Tree model, which achieved 99% accuracy on training and 97.85% on testing while significantly bringing down the FAR to 0.03. Also, the deep learning models based on Ridge regression showed auspicious results while dealing with temporal dependencies in the dataset. Advanced feature selection methods and hyperparameter tuning hence can improve the detection performance of the framework by a good margin and thus make it hugely suitable for protecting IoMT systems against a wide range of cyber threats. This work also significantly contributes to IoMT security enhancement by combining ML and DL for the high demand for advanced and scalable intrusion detection mechanisms in healthcare environments.

Zukaib et al. [28] introduce an intelligent IDS for IoMT, combining meta-learning techniques with federated learning. IoMT is one of the most enabling technologies in contemporary healthcare; it faces serious cybersecurity challenges because these medical data are sensitive. A new dataset, called M-IoT-Env, has been prepared. It is made by mixing IoT and conventional network traffic to be similar to real-world network behavior. This paper mainly focuses on evaluating this framework using the dataset introduced. Experimental results demonstrate that meta-fed IDS provides superior accuracy and low misclassification rates and is effective in known and zero-day cyber threat detection. Federated learning would guarantee privacy in IoMT networks as the operation of the IDS could be conducted in a distributed fashion across various devices without sensitive data being centrally collected. This will go a long way in contributing to IoMT security by proposing a scalable and real-time architecture that is easily adaptable for various healthcare environments. Table 1 summarizes current approaches and performance improvements in intrusion detection and cybersecurity for IoMT systems, detailing methodologies, algorithmic developments, datasets, and key developments. The comparison highlights different approaches—spanning hybrid optimization techniques and meta-learning models to energy-efficient architectures—focusing on their empirical verifications in enhancing detection accuracy, reducing computation overhead, and defending real-time IoMT applications. Figure 2 specifies the trade-offs associated with intrusion detection system feature reduction in IoMT, balancing its advantages—better operational effectiveness, reduced computational load, and energy conservation—against potential detriments, including the danger of loss of key features, the problem of computational scalability, and the loss of real-time performance. Such

Table 1 Summary of approaches and performance improvements in intrusion detection and cybersecurity for IoMT

Study	Approach	Algorithm/Model	Dataset (s)	Key contributions
[20]	Double Particle Swarm Optimization (PSO) for feature selection and hyperparameters	DNN, LSTM, DBN	NSL-KDD, CICIDS2017	Improves NIDS performance by reducing redundant features and optimizing parameters
[19]	Hybrid PCA-GWO for dimensionality reduction and classification	Deep Neural Network (DNN)	NSL-KDD	Enhances IDS in IoMT, focusing on anomaly detection and reduced training time
[22]	Feature selection with PSO and AdaBoost	AdaBoost	NSL-KDD	Detects cyberattacks like DoS and U2R, ensuring high accuracy and recall rates
[21]	Machine learning for real-time stress detection in IoMT environments	KNN, SVM, Bagged Tree, ANN	Custom (IoMT stress data)	Prioritizes data processing in fog computing to reduce cloud load, improving real-time detection in healthcare
[23]	Clustering model for energy efficiency in IoMT	Clustering Model for Medical Applications (CMMA)	Simulation	Optimizes energy consumption and cluster head selection for IoMT devices
[24]	Four-tier cloud-fog architecture for data offloading	Fuzzy logic, MQTT protocols	iFogSim toolkit	Reduces energy consumption and latency in real-time health monitoring systems
[25]	Framework for energy-efficient and secure biosensor networks	Wireless Body Area Networks (WBAN), Kruskal's Algorithm	Simulation	Optimizes communication and security, ensuring energy-efficient data transmission in IoMT
[26]	Intrusion detection with RFE-Decision Tree and Ridge regression	Decision Tree, Ridge Regression, LSTM	WUSTL-EHMS	Combines ML and DL for improved anomaly detection in IoMT, using advanced feature selection methods
[28]	Hybrid meta-learning and federated learning IDS for IoMT	Decision Tree, Random Forest, XGBoost, Meta-Learning	M-IoT-Env (Custom)	Detects known and zero-day attacks in a decentralized manner using federated learning

Fig. 2 Feature reduction in IDS

a study underscores the requirement for a balance between algorithm design to provide optimal detection performance and reduce resource problems in IoMT.

3 Proposed method

This paper presents an optimized feature selection and classification approach to the resource-constrained IDSs in IoMT. We propose a two-stage procedure, mutual information and ensemble-based feature selection, and the RF classifier that provides a high accuracy and low computational cost. The methodology followed for the development of the IDS is presented in this section in detail: feature selection, classifiers involved in ranking features, and the final classification model. Feature selection is an integral part of the proposed method, as IoMT environments are normally bounded by memory and processing power resources. The first step in feature selection is the application of a mutual information filter. Mutual information [29] is a dependency measure between input features and the target variable. Eventually, we chose features with higher MI to ensure that only relevant features remained. This, therefore, allows us to reduce the complexity of a dataset without losing its high capability for prediction. The ‘SelectKBest’ option in our analysis will permit the choice of the best 50% of the input features according to their mutual information score. This greatly reduces the dimensionality, thereby aiding computational efficiency and reducing overfitting. The filtered feature set undergoes further refinement after the MI filter using an ensemble of four models: RF [30], ADA [31], XGB [32], and LGBM [33]. All these models rank features according to their importance in the classification task. The selected models will be able to work with big data, have nonlinear decision boundaries, and be robust in feature selection. Random Forest Classifier: RF can be used to calculate the importance

of features while constructing several trees during training. This gives higher importance to features that consistently contribute to truthful tree classifications.

AdaBoost Classifier: ADA optimized the feature selection by giving higher weights for the features that would correct the misclassified instances at each iteration. It is an adaptive approach that ensures important features in classification are important.

XGBoost Classifier: XGB has been used because of its regularization capabilities and high predictive accuracy. Each feature's importance is determined by how much information it adds to the model, hence underlining important features that play vital roles in making better predictions.

LightGBM Classifier: LGBM is a newer algorithm that uses the histogram technique, and like XGB, it also provides feature importance defined by gain. The LGBM is light and memory-efficient and hence suitable for resource-constrained environments.

After these four models have been run, there will be two different methods in choosing the most important features, namely the Union and Intersection methods.

Union Method: It is a method of collecting the top features of all four models into one. This union method gives relevant features across models by combining the most important features identified by RF, ADA, XGB, and LGBM. It is a pre-defined proportion of the features retained after the MI filtering stage. Each model's selected number of features forms a pre-defined proportion of the total features retained after the MI filtering stage.

Intersection Method: We seek in this step those features consistently ranked high by all four models. First, the interaction of top-ranked features ensures that only those most robust and significant features are retained, which are universally important across all models, for the final classification task. In short, it is through these union and intersection-based methodologies for feature selection that the dataset squeaks to retain a subset of features that would be comprehensive yet relevant and consistent. This two-tier approach ensures significant feature selection without overloading the system with data.

Classification: Following the feature selection above, an optimal feature subset has now been decided on classification forms, the final step. In this context, only the Random Forest classifier is employed as RF has performed well when handling high-dimensional data, is robust against overfitting, and is sufficient for nonlinear decision boundary handling. In the training phase, RF considers an ensemble of several decision trees, and each decision tree is trained on a bootstrap sample of the data. In the case of a classification problem, the output will exactly be the mode of classes predicted by individual trees. The mechanism of majority votes allows RF to be especially suitable for detecting complex patterns, such as IoMT intrusion detection, where several types of attacks can differ subtly. Given the mutual information and ensemble-based ranking methods, the proposed method enables the RF classifier to identify intrusion attempts with different characteristics effectively and efficiently by using the reduced feature set. In addition, fewer features guarantee that the least load in computation is secured, which provides the basis on which the feasibility of the presented approach can be assured for resource-constrained IoMT environments. To test the effectiveness of the proposed method, the NSL-KDD, CICI IoMT dataset 2024, and the WUSTL-EHMS-2020 datasets are used as a benchmark for intrusion detection. The categorical variables are encoded, and the number of features in the dataset is reduced using the MI filter to perform preprocessing. Then, ensemble models would use the final set of

features to train the RF classifier. These models are evaluated based on accuracy, precision, recall, and F1-score. Further, confusion matrices would give insight into the classifier's performance at various attack types: DoS, R2L, and U2R. Memory usage and processing time are monitored to ensure the system remains practical for real-time IoMT intrusion detection. These results prove that the proposed method effectively balances detection accuracy with computational efficiency and thus presents a promising IoMT security solution. An approach to intrusion detection in IoMT environments has been presented for efficient feature selection and robust classification. The approach presented here removes noise from the data through mutual information filtering. It applies ensemble-based feature ranking to ensure the highest possible accuracy while reducing computational demands. Algorithm 1 and Fig. 3 present a hybrid strategy for feature selection for resource-limited intrusion detection in medical IoT (IoMT). The procedure is initiated with preprocessing, during which the labels and the categorical data are encoded. Mutual information filtering maintains the most significant features in the top half. Four ensemble learners—Random Forest, AdaBoost, XGBoost, and LightGBM—are trained later, whose ranking of features is utilized for selecting features. The union strategy captures various high-ranked features by each learner, while the intersection strategy captures the high-priority features in common among all the learners. These subsets are combined into a final set of features. This approach balances computation efficiency and accuracy by applying dimensionality reduction using mutual information and ensemble-based selection for robustness. The union-intersection strategy aligns with prior work in scalable feature selection and holds promise for real-time IoMT security, with the caveat of needing to be empirically tested for performance in dynamic threat scenarios.

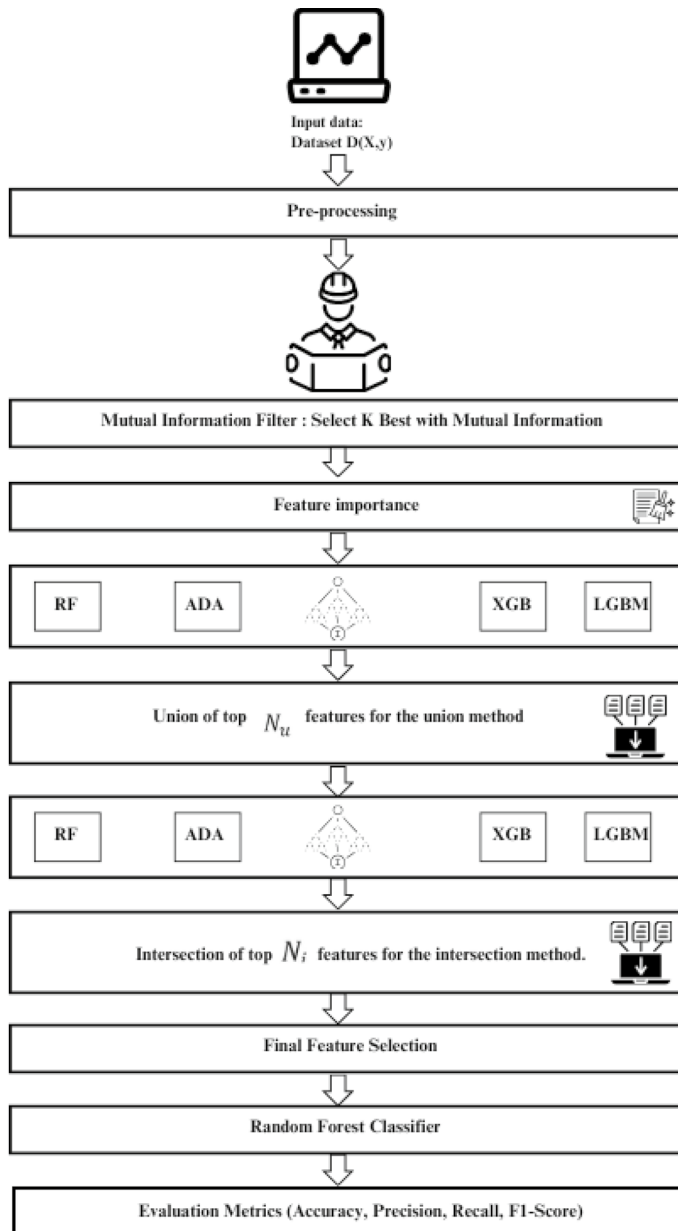


Fig. 3 Proposed method

Algorithm 1 For Feature Selection and Classification in Resource-Constrained IDS for IoMT

Input:

Dataset D with features X and labels y .

Number of top features for union N_u .

Number of top features for intersection N_i .

Output:

$F_{selected}$ Selected features

Performance metrics of the classification model: accuracy, precision, recall, F1-score.

Preprocessing

Load the dataset $D = (X, y)$

Encoding Categories:

Label encode any categorical features in X using label encoding.

Encode target variable y using label encoding

Mutual Information Feature Selection

Mutual Information Filter

Apply Mutual Information (MI) scoring to calculate each feature in X its dependency on the target variable y .

Select the top 50% of features according to the MI scores to form the reduced feature set. X_{MI} .

Ranking Feature Importance by Ensemble Methods
Initialization of Models

Implementation classes for the following models: RF- Random Forest, ADA- AdaBoost, XGB- XGBoost, and LGBM - LightGBM.

Computing Feature Importances

Train each RF, ADA, XGB, and LGBM models on X_{MI} And y .

For every model M , extract the feature importance scores. I_M

Union and Intersection of Features

Union Method

For every model M , pick the top N_u features according to I_M

Find the union of these features among all models to get F_{Union} .

Intersection Method

For each model M , select the top N_i features based on I_M .

Calculate the intersection of those features across all models and have $F_{Intersection}$.

Selection of the final set of features

Combine Results

Combine Union and Intersection sets of features of both methods into a single feature set called the final selected. $F_{selected}$ Feature set.

Algorithm End

3.1 Mutual information

Mutual information (MI) [10] quantifies mutual dependence between two variables. In feature engineering, MI is particularly applicable to choosing the features most informative about the target variable in a machine learning model; slightly more rigorously, for two random variables, $I(X; Y)$ is defined as follows:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (1)$$

Compared to linear correlation metrics, MI provides a relationship in both the linear and nonlinear domains. It is very suitable for complex models. This nonparametric nature allows MI to detect arbitrary dependencies without assuming that certain relationships between variables exist. One could rank features by their relevance via MI before training a machine learning model, reducing the problem's dimensionality,

and often improving model performance. Other methods, such as the Minimum Redundancy Maximum Relevance algorithm, extend this basic approach by viewing both relevance via MI and redundancy between features [10].

3.2 Bagging and boosting feature importance

Bagging and Boosting are the most potent ensemble techniques to improve predictive accuracy and model robustness. While their approach is fundamentally different, both techniques bear significant implications for impurity-based feature importance analysis in the context of model interpretation. Bagging, or Bootstrap Aggregating, is meant to reduce the variance of the models by training several base learners on different subsets of data and averaging out their predictions. Let the Bagging ensemble be described as [34]:

$$\hat{f}(x) = \frac{1}{B} \sum_{i=1}^B \hat{f}_i(x) \quad (2)$$

where $\hat{f}_i(x)$ is the prediction from the i -th model, and B is the number of models. Feature importance in Bagging usually comes from aggregating the importance scores of the base models. Bagging reduces overfitting by regularizing variance, and it gives very stable rankings. Boosting builds the models iteratively, giving higher weights at each iteration to misclassified samples; the weighted sum of weak learners defines the overall model.

We can write the final model [35]:

$$F(x) = \sum_{t=1}^T \alpha_t h_t(x) \quad (3)$$

where $h_t(x)$ is the weak learner at iteration t and is the weight for that learner. Boosting scale features related to instances that are challenging to classify. Therefore, it dynamically changes the importance of features from one iteration to another. Both methods have significant consequences in the field of feature selection; Bagging offers reliable rankings of features because it resists noise, while Boosting concentrates on those features responsible for correcting errors, which yields detailed importance scores [36]. Figure 4 illustrates key problems in IoMT IDS performance, including resource limitations, feature optimization, data management, and security threats. Resource limitations in limited processing power and memory are tackled by strategically removing features with redundancy while maintaining important information. The study identifies the relationship between real-time monitoring demands and data quantity problems, while security threats are centered around device tampering and data leakage. The framework is based on existing research in lightweight IDS implementations in IoMT systems with limited resources.

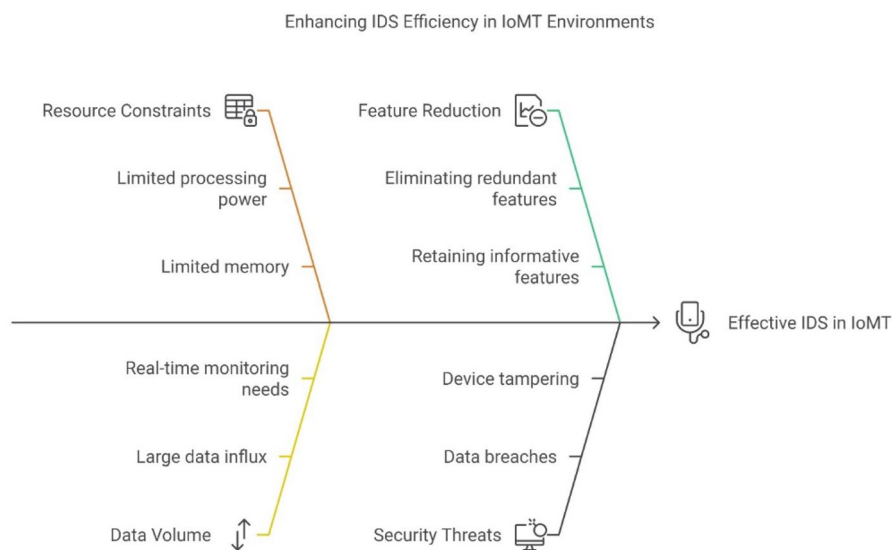


Fig. 4 Enhancing IDS efficiency in IoMT

4 Dataset description and experiment environment

4.1 Experiment environment

We have used The Intel® Core™ i5-12400 Processor, a six-core processor with a base clock of 2.50 GHz, and can run all processor cores at maximum turbo frequencies up to 4.40 GHz. It is supported by 32 GB of DDR4 RAM and hence was computationally adequate for feature selection processes and the training of models. Data preprocessing, reduction, and performance analysis for different models were done in Python, taking advantage of packages like pandas for data manipulation, scikit-learn for machine learning, feature reduction methodologies, and matplotlib and seaborn for visualization. This inherently means one thing: Under the hood, it is very computationally efficient, even at moderate and large datasets; it was perfectly fine to run all types of algorithms. The configuration of balanced hardware and software processing power with memory capacity was built upon for the best performance of machine learning experiments.

4.2 CICI IoMT dataset 2024

CICIoMT-2024 dataset [37], the IoMT dataset, developed by the Canadian Institute for Cybersecurity at the University of New Brunswick, is a comprehensive dataset for addressing security vulnerabilities in the fast-growing domain of IoMT. Since IoMT technologies would be deeply integrated into general healthcare systems, ensuring security and privacy concerning medical

devices and their communications is paramount. This dataset is designed to enable research into this domain, specifically the capability for detecting and mitigating different kinds of cyberattacks against IoMT infrastructures. IoMT comprises various medical devices, software, and services that interrelate through sharing data over the internet toward better healthcare delivery outcomes. These include wearable devices, patient-worn monitoring systems, intelligent hospital beds, and automated medication dispensers. This, however, puts them in the firing line of cyberattacks due to their usually anemic computational resources and less-than-perfect security hierarchies. The repercussions can also be disastrous, such as manipulated sensitive medical data, a necessary service brought down, or leakage of personal health information. CICIoMT2024 is based on network traffic emanating from 40 IoMT devices, of which 25 are real physical devices, and 15 are simulating standard devices usually encountered in healthcare settings. Several devices rely on different communication protocols, specifically Wi-Fi, MQTT-Message Queuing Telemetry Transport-and Bluetooth, to reflect the heterogeneous nature typical of real IoMT deployments. The dataset summary is as follows, and it is divided into two significant directories. The first one is called Bluetooth Traffic Directory, in which the traffic will emanate from devices communicating over Bluetooth, and considering that many wearables—like fitness trackers or heart rate monitors—are connected via Bluetooth, this part of the dataset represents a wide range of activities, both legitimate and malicious, specific to this protocol. Another essential characteristic of the CICIoMT2024 dataset is that 18 cyber-attacks were simulated to realize IoMT vulnerabilities. These major kinds of attacks fall into several categories: Distributed Denial-of-Service (DDoS): An overloaded target device or network is flooded with tons of traffic, making it unavailable for a legitimate user. Such attacks cause fatal impacts in health sectors where device unavailability is sometimes critical. DoS: Similar to DDoS, DoS tries to bring down a node or a service; it generally emanates from one source. The dataset simulates various types of DoS attacks. Its main applications are within machine learning for developing intrusion detection and anomaly detection algorithms. Due to the attacks' diversity, including actual and simulated devices enables a wide range of testing scenarios. Besides cybersecurity research, this dataset can also be helpful in network analysis, traffic classification, and protocol optimization studies. This is furthered by using real IoMT devices to ensure the captured data are easily representative of the actual behavior of real devices in operational, clinical healthcare environments. The CICIoMT2024 dataset presented here represents a critical development in IoMT security research. The dataset provides a comprehensive and varied repository of traffic data covering not only legitimate activities but also malicious ones; it opens the door for researchers to take on key cybersecurity challenges in the healthcare sector. As IoMT is going to be adopted more and more in the future, such datasets as CICIoMT2024 will indeed prove to be a milestone in enhancing the security and reliability of these integral technologies. The CICIoMT2024 class distribution is highly skewed among different traffic classes. The data distribution pattern, as observed in Fig. 5, is long-tailed, with very sparse classes having very few instances, while other classes have a very high number of instances.

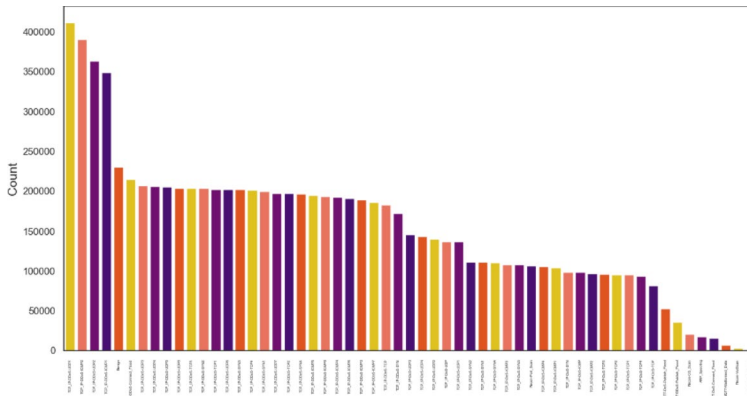


Fig. 5 CICIoMT2024 class distribution

The most prominent classes have approximately 350,000–400,000 instances, while the sparse classes have less than 50,000 instances. The intrinsic skewness in the classes is a significant hindrance to the model’s training, and great care is needed in the choice of balancing techniques to ensure excellent classification performance in each traffic class.

4.3 The NSL-KDD dataset

The NSL-KDD dataset [38] was developed to evaluate the IDS and is one of the most popular datasets in recent cybersecurity research, especially when developing and testing machine learning models. The different attacks are categorized into four main categories using the NSL-KDD dataset. Denial-of-Service (DoS): This class of attack concentrates on confronting a system through which there would not be any resource access available to legitimate users. Typical examples are SYN flood and Smurf attacks. Probe: In this type of attack, probe attacks monitor information about a system to detect possible vulnerabilities. This may be in port scanning or something else to map out the network. U2R: The attacker breaks into a system and then uses a vulnerability to gain root access from a standard user account in these attacks. The dataset structure separated into training and testing subsets enables the model to generalize from already known types of attacks into novel or evolving threats. In addition, a broad representation of attack types gives a complex testing playground for security heuristics. The NSL-KDD dataset plays a crucial role in developing intrusion detection systems. It improves upon the original KDD’99 dataset in several ways and, thus, can be a much more reliable helper for researchers in developing machine learning models for intrusion detection, focusing on existing cyber-attacks and new ones. It avoids the redundancy and imbalance issues regarding the earlier version and provides a more realistic and challenging environment while testing the efficiency of intrusion detection techniques. The NSL-KDD data show a highly skewed distribution in different classes of attacks, as can be seen in Fig. 6. Normal traffic comprises the majority, with approximately 67,000 instances, while

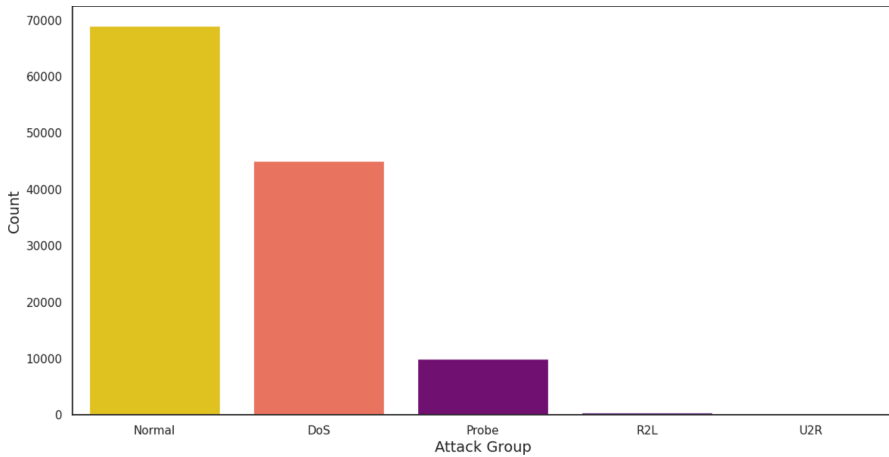


Fig. 6 NSL-KDD dataset class distribution

DoS attacks comprise approximately 45,000 instances. Probe attacks are a much smaller group, while R2L and U2R attacks are underrepresented, with very few instances. The skew in the distribution reflects real-world network traffic behavior in which the normal behavior and certain classes of attacks happen more than others, although it does the work of developing.

4.4 The WUSTL-EHMS-2020 dataset

The WUSTL-EHMS-2020 [15, 39] is a cybersecurity resource created to identify and deal with various vulnerabilities within an IoMT setting through a simulated attack in a healthcare environment. It utilizes an integrated testbed of patient biometric data and network traffic in real time, where this dataset focuses on two critical attack types, mainly man-in-the-middle spoofing and data injection. These attacks try to discard the integrity and confidentiality of sensitive medical data, making this dataset quite helpful in developing intrusion detection systems in healthcare environments [15]. The dataset offers more than 16,000 labeled samples, divided into standard and attack scenarios. Sample distribution includes 87.5% regular and 12.5% attack traffic, thus providing a good mix of benign and malicious data. These records have 44 different features involving network flow metrics and biometric data of patients. The real-time biometric data of patients are one critical attribute engaged in this data, making it unique compared to other available IoTs or IoMT datasets and offering an extensive, profound view of how cyberattacks can influence healthcare systems. The most significant attributes included in this dataset are network performance measurement data, such as packet count, average packet size, and inter-arrival times. These are joined with patient-specific data, including heart rate and blood oxygen levels, which can allow the simulation of the effect on real-time medical monitoring if there is disruption on the network or data tampering. The MITM spoofing attack in the dataset provides a scenario wherein the intruder

accesses unauthorized communication between IoMT devices and the server or medical application. The attacker intercepts, alters, or injects data to mislead the healthcare system and thus delay the treatment or wrong medical response. In a Data Injection Attack, an adversary injects malicious or false data into the communication streams. Such an attack would be targeted to corrupt the patient data in transit so that the patient may appear in a worse state than they are or hide/camouflage the signs of critical emergencies. The dataset will find broad applications in developing and evaluating Intrusion Detection Systems IDSs using machine learning for detecting and mitigating IoMT cyber threats. Combining biometric data with network data is handy when devising an IDS capable of detecting technical anomalies in network traffic and physiological ones in patient data. Moreover, it could be much more relevant to the works related to anomaly detection, where researchers strive to find some unusual patterns that show traces of an ongoing cyber-attack or system failure. The WUSTL-EHMS-2020 dataset combines accurate patient data with simulated attack scenarios, and based on this, an even more exciting evaluation of such attacks' influence on the security of patient care and the reliability of medical devices could be done. Real-time allows detailed studies of latency problems, data integrity, and medical personnel's response to false alerts because of spoofing or injection attacks. The WUSTL-EHMS-2020 dataset illustrates the heavy class skew present in real-world healthcare monitoring systems, as shown in Fig. 7. Normal behavior is the most prevalent, with approximately 14,000 instances, while Spoofing attacks and Data Alteration attacks are much rarer, with approximately 2,000 instances each. This distribution pattern is characteristic of healthcare monitoring situations where normal operations vastly outnumber security events.

4.5 Feature reduction

Our feature reduction contains two stages: union-based reduction and intersection-based reduction. Network-related features may include protocols such as UDP, TCP, ICMP, rates, durations, and flags, among many others, present in different protocols

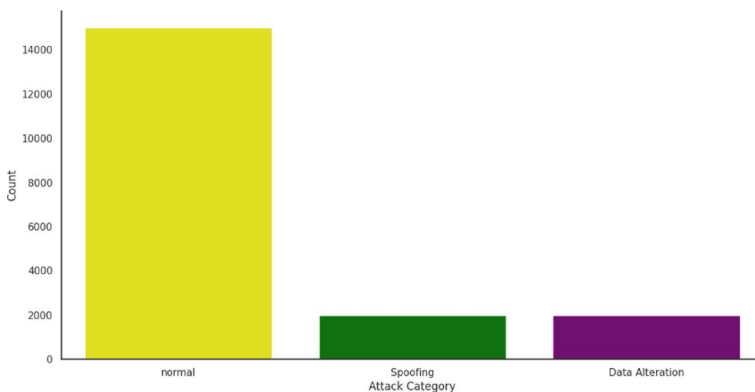


Fig. 7 WUSTL-EHMS-2020 class distribution

of the CICIoMT2024 dataset. Therefore, it depicts various attributes in building network anomaly detection. The union-based feature selection: Most of the noted machine learning models, such as RF, XGB, LGB, and ADA, use a union method whereby their chosen features are combined in one collective set. This ensures that each feature selected by at least one model remains within that final set. The features maintained by this union include protocol types like UDP, ICMP, and TCP; different flag numbers such as SYN and FIN; sizes and rates; and statistical measurements that include mean representation. Intersection-based Feature Selection: Only those features consistently selected by all the models are retained. This ensures a final, more conservative, and reduced feature set. While the dimensionality is significantly reduced, it retains the most significant predictors for anomaly detection. The union-based feature selection keeps a wide range of features for source and destination attributes. This provides a more workable feature set, keeping in mind prime factors above the average rating of most models. Intersection-based Feature Selection: The intersection approach refers to critical health metrics such as heart, pulse_Rate, and resp_Rate, as well as essential network features like SrcMac, Rate, and DstJitter. This drastically reduces the feature set to handpick the most dependable markers of health status and stability across communications. The process of feature reduction in WUSTL-EHMS-2020, NSL-KDD, and CICIoMT2024 datasets employs the same systematic process, as illustrated in Figs. 8, 9, and 10. The process begins with the features in Layer 1, proceeding through probing and testing phases in order to select important features. The process in the workflow involves discrimination, resulting in the selection of features in Layer 2 and leading to the final feature subset. The same standard pipeline is applied in order to have the same selection of features in the three datasets, enabling dimensional reduction with the preservation of the discriminative power of the features selected. The DT-PBFS approach integrates decision tree-based selection with performance-based filtering, maximizing the feature space to enhance classification efficiency.

Feature reduction after μ in the three datasets shows extensive dimensionality reduction through union and intersection-based reductions, as shown in Fig. 11. In CICIoMT2024, the 45 original features were reduced to 40 through union-based

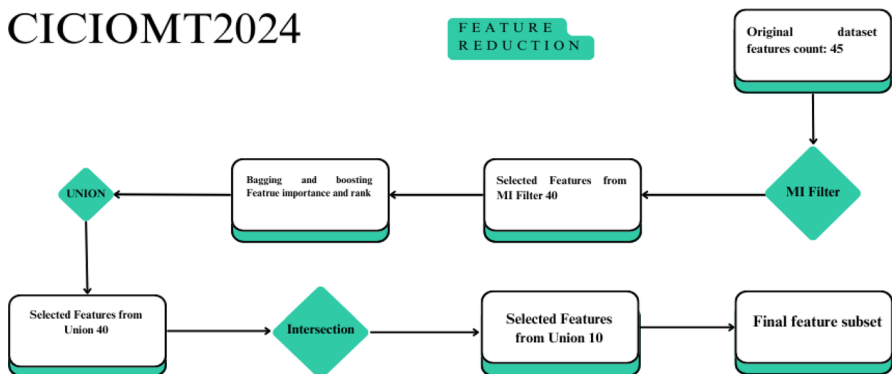


Fig. 8 Feature reduction process for CICIoMT-2024

NSLKDD

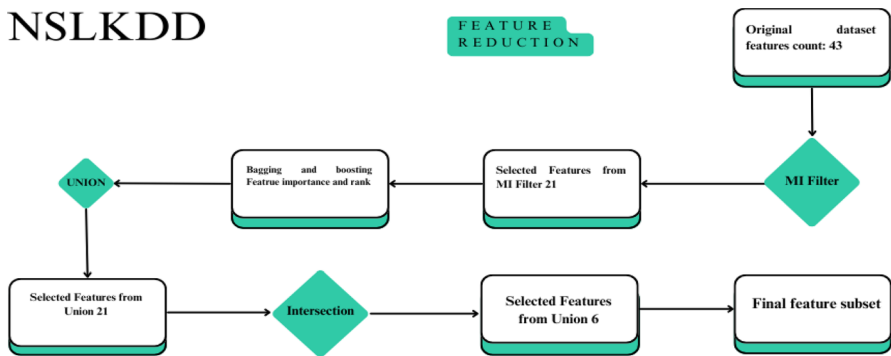


Fig. 9 Feature reduction process for NSL-KDD

WUSTL-EHMS 2020

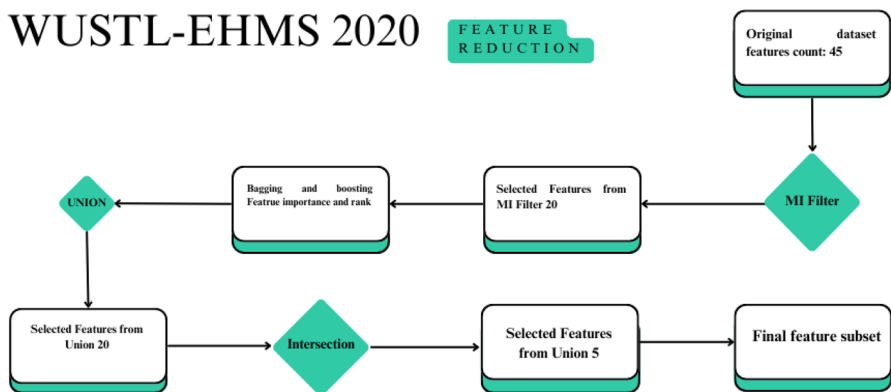


Fig. 10 Feature reduction process for WUSTL-EHMS-2020

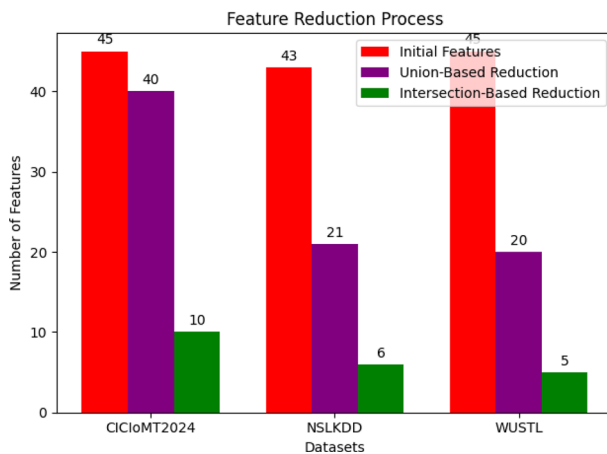


Fig. 11 Feature reduction after MU

reduction and compressed to 10 features through intersection-based reduction. The NSL-KDD dataset originally had 43 features, which were reduced to 21 features and later to 6 features. Similarly, the WUSTL-EHMS dataset originally had 41 features, which were reduced to 20 features through union-based reduction and finally to 5 features through intersection-based reduction. This progressive feature reduction effectively minimizes the computational complexity while keeping the inherent discriminative features of each dataset.

Based on Table 2, the last characteristic features capture the most important network features in datasets. CICIoMT2024 preserves 10 features emphasizing protocols and traffic parameters, and WUSTL preserves five features emphasizing communication between devices. In comparison, NSL-KDD preserves six features emphasizing host behavior and traffic data.

5 Experimental result

5.1 CIC IoMT dataset 2024 on original dataset with random forest

In Tables 3 and 4, the following performance reports are an overall analysis of the Random Forest model classification performance conducted on the CIC IoMT dataset 2024. The dataset contains many network anomalies, which span most of the varieties representing DDoS and DoS attacks, along with benign traffic. For each of several attack categories, the model's performance is evaluated. Precision, recall, and F1-score will be the primary evaluation metrics. The reason is that these provide insight into the ability of the classifier to identify each class in the dataset, and precision gives an idea of how many of the predicted positive cases were correct. The precision values in most attack types are high, mostly more significant than 90%, achieving even perfect precision in many classes like the MQTT-DDoS and TCP_IP-DDoS attacks. More similar trends can be seen in the recall. For benign traffic, the recall values were 100%, in that it ideally classified 100% of nonmalicious traffic. For other classes, this slightly drops: "Recon-OS_Scan" has a recall of 0.73, hence showing that the model has missed some true positives. Thus, this describes the general performance of each class. Indeed, the Random Forest model performed exceptionally well for most classes, having an F1-score close to or precisely 1.00, including those DDoS- and DoS-related classes. These include exceptions like "Recon-Ping_Sweep," which has an F1-score of 0.82, implying further room for improvement in such categories. MQTT-DDoS and MQTT-DoS

Table 2 Final feature set

Dataset	Selected features
CICIoMT2024	syn_count, IAT, ICMP, rst_count, fin_count, UDP, Protocol Type, Min, Magnitude, Tot size
WUSTL	SrcMac, DIntPkt, SrcLoad, SrcJitter, DstJitter
NSLKDD	dst_host_serror_rate, dst_host_diff_srv_rate, level, dst_bytes, src_bytes, count

Table 3 CIC IoMT dataset 2024 on original dataset with Random Forest

Class	Precision	Recall	F1-Score	Support
ARP_Spoofing	0.92	0.87	0.89	4407
Benign	0.98	1.00	0.99	57,366
MQTT-DDoS-Connect_Flood	1.00	1.00	1.00	53,926
MQTT-DDoS-Publish_Flood	1.00	1.00	1.00	9074
MQTT-DoS-Connect_Flood	1.00	1.00	1.00	4047
MQTT-DoS-Publish_Flood	1.00	1.00	1.00	13,001
MQTT-Malformed_Data	0.95	0.87	0.91	1741
Recon-OS_Scan	0.92	0.73	0.82	5212
Recon-Ping_Sweep	0.91	0.75	0.82	233
Recon-Port_Scan	0.95	0.98	0.97	26,722
Recon-VulScan	0.88	0.71	0.78	798
TCP_IP-DDoS-ICMP1	0.96	0.99	0.97	87,349
TCP_IP-DDoS-ICMP2	0.96	0.97	0.97	97,222
TCP_IP-DDoS-ICMP3	0.97	0.91	0.94	47,490
TCP_IP-DDoS-ICMP4	0.93	0.94	0.93	48,520
TCP_IP-DDoS-ICMP5	0.96	0.94	0.95	48,743
TCP_IP-DDoS-ICMP6	0.97	0.97	0.97	47,593
TCP_IP-DDoS-ICMP7	0.95	0.99	0.97	46,759
TCP_IP-DDoS-ICMP8	1.00	0.96	0.98	48,328
TCP_IP-DDoS-SYN	0.89	0.97	0.93	43,153
TCP_IP-DDoS-SYN1	0.96	0.99	0.98	49,678
TCP_IP-DDoS-SYN2	0.95	0.96	0.96	50,832
TCP_IP-DDoS-SYN3	0.96	0.86	0.90	50,326
TCP_IP-DDoS-SYN4	0.99	1.00	0.99	48,745
TCP_IP-DDoS-TCP	0.98	0.96	0.97	45,625
TCP_IP-DDoS-TCP1	0.93	0.92	0.93	50,648
TCP_IP-DDoS-TCP2	0.93	0.94	0.93	49,398
TCP_IP-DDoS-TCP3	0.97	0.98	0.97	51,149
TCP_IP-DDoS-TCP4	1.00	0.98	0.99	50,241
TCP_IP-DDoS-UDP1	0.94	0.95	0.94	103,243
TCP_IP-DDoS-UDP2	0.95	0.91	0.93	91,505
TCP_IP-DDoS-UDP3	0.91	0.98	0.94	51,704
TCP_IP-DDoS-UDP4	0.92	0.80	0.85	51,569
TCP_IP-DDoS-UDP5	0.86	0.89	0.88	51,422
TCP_IP-DDoS-UDP6	0.89	0.90	0.89	50,372
TCP_IP-DDoS-UDP7	0.86	0.92	0.89	49,261
TCP_IP-DDoS-UDP8	0.96	0.95	0.95	51,049
TCP_IP-DoS-ICMP	0.96	0.98	0.97	24,695
TCP_IP-DoS-ICMP1	0.93	0.89	0.91	26,292
TCP_IP-DoS-ICMP2	0.92	0.95	0.93	24,068
TCP_IP-DoS-ICMP3	0.96	0.94	0.95	27,091
TCP_IP-DoS-ICMP4	0.94	0.95	0.95	26,568

Table 3 (continued)

Class	Precision	Recall	F1-Score	Support
TCP_IP-DoS-SYN	0.89	0.96	0.93	24,693
TCP_IP-DoS-SYN1	0.93	0.97	0.95	27,754
TCP_IP-DoS-SYN2	0.96	0.99	0.97	28,201
TCP_IP-DoS-SYN3	0.96	0.87	0.91	27,158
TCP_IP-DoS-SYN4	1.00	0.94	0.97	27,456
TCP_IP-DoS-TCP	0.99	0.97	0.98	20,433
TCP_IP-DoS-TCP1	0.99	1.00	0.99	23,698
TCP_IP-DoS-TCP2	1.00	0.96	0.98	24,007
TCP_IP-DoS-TCP3	0.94	0.99	0.97	24,031
TCP_IP-DoS-TCP4	0.97	0.96	0.97	23,449
TCP_IP-DoS-UDP	0.99	0.97	0.98	34,319
TCP_IP-DoS-UDP1	0.95	0.87	0.91	34,056
TCP_IP-DoS-UDP2	0.94	0.96	0.95	35,047
TCP_IP-DoS-UDP3	0.96	0.99	0.98	36,495
TCP_IP-DoS-UDP4	0.94	0.98	0.96	35,792
Macro-avg	0.95	0.94	0.94	2,193,754
Weighted avg	0.95	0.95	0.95	2,193,754

Accuracy: 0.9497723992753974

Table 4 CIC IoMT dataset 2024 report on original dataset

Metric	Value
Time taken	1878.2257182598114 s
Memory usage at the end	23,962.33203125 MB
Memory used	23,736.89453125 MB
CPU usage	2.2%

Floods: Perfectly classified, precision, and recall are 1.00, providing F1-score values 1.00. The model uses the random forest to differentiate between attacks, benign traffic, or other classes without a hitch. TCP_IP-DDoS-ICMP1, TCP_IP-DDoS-ICMP2, and TCP_IP-DDoS-SYN: Considering a class of ICMP and SYN flooding attacks, this also demonstrates near-perfect precision and recall. This model is doing quite well, particularly for detecting and classifying these traffic patterns, with most variants having precision and recall greater than 0.95. Recon-OS Scan: 0.82, Recon-Ping Sweep: 0.82: These are classes in reconnaissance-based attacks that have lowered performance. They have lower recall than other classes, which might indicate that this model does not detect specific variants of reconnaissance traffic. Execution Time: It took 1878.23 s for the training and evaluation, which is quite assumable concerning a computational cost when considering the complexity of the dataset. Memory Usage: The memory consumed at the end by the model was around 23,962.33 MB. Thus, most memory consumption can be kept strictly within

training and evaluation. CPU usage stayed as low as 2.2%, meaning the bottleneck is likely memory consumption rather than capacity. Indeed, the Random Forest model shows perfect class balancing regarding different attack types and precision and recall. It is most successful in detecting MQTT and TCP_IP-based DDoS and DoS Attacks, for which the classification performance was nearly perfect. That is important for practical usage since the type of attacks in question has dramatically grown in IoT setups. The model has specific areas for improvement in recognizing reconnaissance activities since lower performance metrics are seen in classes like Recon-OS Scan and Recon-Ping Sweep. It follows through with the fact that the Random Forest model, as applied to the CIC IoMT dataset, gives an outstanding performance in most of the attack classes. However, there are a few areas where the model can be holistically improved. High accuracy, complemented by high precision and recall against critical attack types, makes the Random Forest a widely valuable tool in detecting and mitigating of network- based attacks. Table 3 shows Random Forest performance on the CIC IoMT 2024 dataset shows strong results with 95% overall accuracy. Most attack classes achieve high precision (>0.90) and recall (>0.90), with MQTT-DDoS floods showing perfect detection (1.00). There is lower performance in reconnaissance attacks (Recon-VulScan: $F1=0.78$) and some UDP-based DDoS attacks (TCP_IP-DDoS-UDP5: $F1=0.88$). Benign traffic classification remains highly accurate ($F1=0.99$).

Based on Table 4 CIC IoMT 2024 data processing metrics show an execution time of approximately 1878 s (≈ 31 min), maximum memory usage of 23.96 GB, 23.74 GB of memory, and minimum CPU usage of 2.2%.

5.2 CIC IoMT dataset 2024 on proposed method with random forest

Tables 5 and 6 show the CIC IoMT dataset 2024 classification report with proposed method for detecting and classifying various network traffic anomalies. The precision values are very high in most of the attack categories; for MQTT-DDoS and MQTT-DoS classes, perfect precision was achieved by the model, which is 1.00. MQTT-DDoS and MQTT-DoS achieve perfect recall, indicating excellent performance by the model in catching those malicious flows. Classes such as MQTT-DDoS-Publish_Flood reach an extremely high F1-score of 1.00, while TCP_IP-DDoS-SYN4 is 0.99. That is very important because such types of DDoS attacks undoubtedly will shut down the network functionality within the IoT environment. Other classes performing well were TCP_IP-DDoS-ICMP and TCP_IP-DDoS-UDP, with F1-scores of 0.90 to 0.97, signifying that the model performed exceptionally well in classifying the attacks using both ICMP and UDP. Recon-OS_Scan and Recon-VulScan: The poorest performance, with an F1-score of 0.70, comes from these reconnaissance-related attack classes. These results indicate that this model could better detect reconnaissance attempts, which are crucial in network security since they always precede a critical attack. Though this takes 604.50 s to execute, as opposed to 1878.23 s on the original dataset, this represents a massive improvement in computational efficiency. The reduced time may be significant in such applications where real-time or near-real-time network intrusion detection systems

Table 5 Classification report of CIC IoMT dataset 2024 on proposed method with Random Forest

Class	Precision	Recall	F1-Score	Support
ARP_Spoofing	0.79	0.73	0.76	4407
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Table 5 (continued)

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TCP_IP-DoS-SYN2	0.95	0.99	0.97	28,201
TCP_IP-DoS-SYN3	0.94	0.88	0.91	27,158
TCP_IP-DoS-SYN4	0.99	0.94	0.96	27,456
TCP_IP-DoS-TCP	1.00	0.95	0.97	20,433
TCP_IP-DoS-TCP1	0.98	1.00	0.99	23,698
TCP_IP-DoS-TCP2	1.00	0.94	0.97	24,007
TCP_IP-DoS-TCP3	0.91	1.00	0.95	24,031
TCP_IP-DoS-TCP4	0.96	0.95	0.95	23,449
TCP_IP-DoS-UDP	1.00	0.98	0.99	34,319
TCP_IP-DoS-UDP1	0.99	0.85	0.92	34,056
TCP_IP-DoS-UDP2	0.93	0.98	0.96	35,047
TCP_IP-DoS-UDP3	0.96	1.00	0.98	36,495
TCP_IP-DoS-UDP4	0.93	0.99	0.96	35,792
Macro-avg	0.93	0.92	0.92	2,193,754
Weighted avg	0.94	0.93	0.93	2,193,754

Accuracy: 0.9348044493594085

Table 6 CIC IoMT dataset 2024 on proposed method

Metric	Value
Time taken	606.4958882331848 s
Memory usage at the end	11,489.6796875 MB
Memory used	11,264.76953125 MB
CPU usage	6.5%

are necessary. Execution Time: The proposed method requires far less memory: 11,489.68 MB at the end of its execution compared to 23,962.33 MB for the original model. Therefore, this makes the proposed method more resource-friendly and suitable for deployment even in resource-constrained computational environments such as IoT devices and networks. CPU Usage: The proposed has more load on the CPU, 6.5% against 2.2% of the original model. Though this might be heavier, it is still relatively low and presents no real bottleneck in most environments. Generally, the performance was excellent concerning classifying various types of attacks, mainly those DDoS or DoS types, using protocols such as MQTT, TCP, and UDP. Indeed, they performed very well regarding the classification, being almost perfect in terms of precision and recall for several important classes of attacks. This high performance can be crucial in IoT environments, given that timely and effective detection of network intrusion attempts is highly decisive. Computationally, the proposed technique significantly reduces execution time and memory usage, making

Table 7 Classification report on original dataset with Random Forest

Class	Precision	Recall	F1-Score	Support
DoS	0.96	0.86	0.91	6904
Normal	0.66	0.97	0.79	9711
Probe	0.80	0.62	0.70	2634
R2L	0.68	0.01	0.01	2524
U2R	0.73	0.05	0.09	767
Macro-avg	0.77	0.50	0.50	22,540
Weighted avg	0.77	0.76	0.70	22,540

Accuracy: 0.7569653948535936

Table 8 Classification metrics on original dataset with Random Forest

Metric	Value
Time taken	6.020453691482544 s
Memory usage at the end	351.57421875 MB
Memory used	126.578125 MB
CPU usage	2.7%

it more suitable for real-time implementation. This results in a relatively small loss in accuracy. At the same time, the gains to be had in computational resource savings make this an attractive approach for resource-constrained hardware environments. The proposed approach hence represents a practical scheme for detecting most types of attacks; it is also much more computationally efficient than the original model.

5.3 The NSL-KDD on original dataset with random forest

Tables 7 and 8 present the original dataset's Random Forest algorithm classification report. The model performance in terms of different classes of network traffic includes Denial-of-Service (DoS), probe, remote-to-local (R2L), and user-to-root (U2R) attacks. The following explains essential metrics for precision, recall, and F1-score for each class in the analysis with the general computational metrics. The class DoS has the highest precision score, 0.96, which indicates high model capability in correctly identifying instances of Denial-of-Service attacks. In contrast, R2L and U2R are subtle attack classes with precision values of 0.68 and 0.73, respectively. The model had a good recall for regular traffic, 0.97, meaning nearly all normal traffic was classified correctly. The recall was very low in R2L 0.01 and U2R 0.05 cases, which says detecting this attack using the model is challenging. The lower recall in the instances of R2L and U2R reveals many false negatives when undetected attacks reach a very high level. A balanced F1-score: The F1-score is the balance between precision and recall. The DoS class had the highest F1-score at 0.91, showing an excellent output on the model's identifying and detecting such an attack. The F1-score also revealed that the regular class was good in classification

but imperfect. The probe class had a reasonably good F1-score of 0.70, but the R2L and U2R classes had shallow F1-scores, at 0.01 and 0.09, respectively. This means extremely poor detection against these more sophisticated kinds of attack types. DoS Attacks: The model performed well in detecting the DoS attacks, with a precision of 0.96 and an F1-score of 0.91, showing the model's strength in detecting this common network attack. The precision on all classes is averaged at 0.77 on the macro-average, which means the model can quite well accommodate all classes regarding precision on average. The overall recall of 0.50 still shows that the model is seriously handicapped in detecting specific attack classes, so much more seriously for R2L and U2R. Memory Consumption: After the run, the average usage of the model was 351.57 MB, and the highest during the run was 126.58 MB. The Random Forest model does well in classifying DoS attacks and regular network traffic, hence the high precision, recall, and F1-scores for these two classes. These traffic classes dominate the dataset; therefore, the overall high accuracy is 75.7%.

Figure 12 shows the NSL-KDD confusion matrix shows strong classification for DoS (5,918 accurate) and normal (9,448 accurate) classes. The probe attacks perform moderately (1,642 accurate), while R2L and U2R classes perform poorly with more misclassifications, with R2L being confused primarily with normal traffic.

5.4 The NSL-KDD dataset on proposed method with random forest

The classification report for the NSL-KDD dataset using a Random Forest algorithm in the proposed method is given in Tables 9 and 10; these tables show a comprehensive performance analysis of the model's ability to classify network traffic into several attack categories: DoS, normal, probe, remote-to-local attack (R2L), and user-to-root attack (U2R). The DoS class had the highest precision of

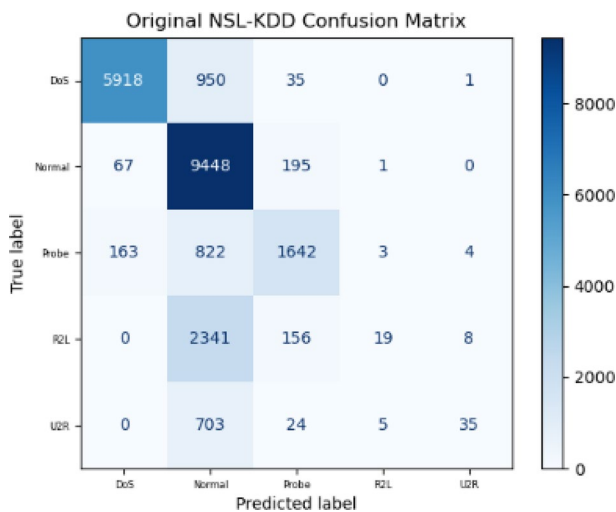


Fig. 12 Confusion matrix

Table 9 Classification report of NSL-KDD dataset on proposed method with Random Forest

Class	Precision	Recall	F1-Score	Support
DoS	0.96	0.82	0.89	6904
Normal	0.69	0.97	0.81	9711
Probe	0.64	0.64	0.64	2634
R2L	0.56	0.04	0.07	2524
U2R	0.41	0.06	0.10	767
Macro-avg	0.65	0.51	0.50	22,540
Weighted avg	0.74	0.75	0.70	22,540

Accuracy: 0.7516858917480036

Table 10 Classification metrics of NSL-KDD dataset on proposed method with Random Forest

Metric	Value
Time taken	4.051710844039917 s
Memory usage at the end	273.265625 MB
Memory used	47.92578125 MB
CPU usage	1.7%

0.96, indicating that the model was quite effective in correctly identifying instances of Denial-of-Service attack classes. The Recall was particularly strong for regular traffic, standing at 0.97; hence, most regular traffic cases were correctly classified. The DoS class scored highest, 0.89, reflecting the model's consistency in identifying and detecting those attacks. For regular traffic, though, the F1-score of 0.81 also reflects solid classification. This model can perform well in identifying DOS attacks with a precision of 0.96 and an F1-score of 0.89, hence fairly representing good capability for detecting overt network disruptions. Regular Traffic: This model performs well in finding regular traffic with a high Recall of 0.97 and an F1-score of 0.81. That implies that regular traffic is barely misclassified, while some attack traffic may be labeled normal, which is explained by the lower precision of 0.69. Probe Attack: In the probe attack class, the F1-score is 0.64-moderate performance, with a good balance between accuracy and Recall, while describing the poor performance in detecting this attack class. Execution Time: The method proposed in this paper presented a low execution time, equal to 4.05 s, which assured excellent computational efficiency, which becomes more and more relevant for real-time intrusion detection systems. Memory Usage: The memory used at the termination point was 273.27 MB, whereas the maximum usage during execution was 47.93 MB. This indicates that when running, the model requires minimal computational resources. CPU Usage: This is further augmented by CPU usage, which is just 1.7%. The above low consumption of resources allows a model to be implemented in real-time systems without affecting performance very much. The proposed Random Forest-based method is suitable for detecting DoS attacks and regular traffic, which shows high precision and recall for these particular classes. The significant classification performance difference within the classes can be attributed to the

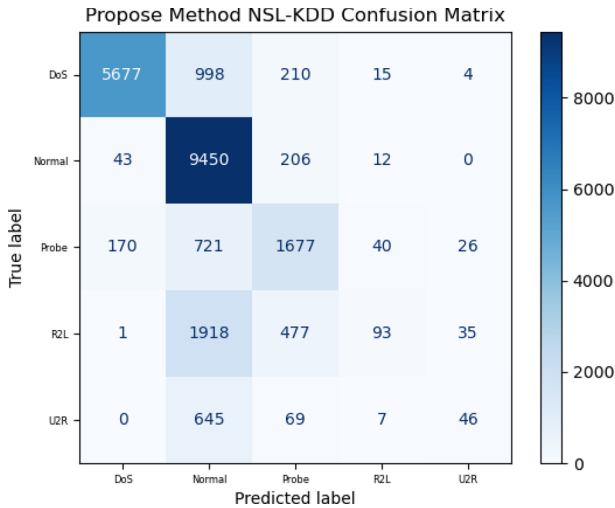


Fig. 13 Confusion matrix

Table 11 Classification report of WUSTL-EHMS-2020 on original dataset with Random Forest

Class	Precision	Recall	F1-Score	Support
Data Alteration	1.00	1.00	1.00	249
Spoofing	1.00	1.00	1.00	288
Normal	1.00	1.00	1.00	3543
Macro-avg	1.00	1.00	1.00	4080
Weighted avg	1.00	1.00	1.00	4080

Accuracy: 0.9997549019607843

imbalance in the NSL-KDD dataset, where the DoS and normal traffic are in an overwhelming majority. At the same time, R2L and U2R form the minority.

Figure 13 shows that the confusion matrix for the proposed approach for NSL-KDD DoS (5677 correct) and normal (9450 correct) with strong detection. The probe attacks were boosted slightly (1677 correct), with R2L (93 correct), and U2R (46 correct) showing slight improvements in the classification rate in comparison with the baseline model.

5.5 The WUSTL-EHMS-2020 on original dataset with random forest

Tables 11 and 12 contain the classification report from a Random Forest model that performed nearly flawlessly on all classes in the WUSTL-EHMS-2020 dataset. The high recall suggests that none of the actual instances of Data Alteration, Spoofing, or regular traffic go undetected. F1-score: Because precision and recall are perfect, the F1-score, too, is identical at 1.00 across. That indicates that the model balances false positives and false negatives exceptionally well. Data alteration: The model has

Table 12 Classification metrics of WUSTL-EHMS-2020 on original dataset with Random Forest

Metric	Value
Time taken	1.3646366596221924 s
Memory usage at the end	264.1171875 MB
Memory used	38.7265625 MB
CPU usage	2.0%

ideally identified the situations of data alteration at precision, recall, and F1-score at 1.00. Spoofing: Similar to the class of data alteration, spoofing attacks are ideally detected; not a single instance is misclassified. Regular Traffic: Normal traffic also shows perfect classification, meaning no false positives or negatives occur. The Random Forest model applied on the WUSTL-EHMS-2020 dataset yields an outstanding performance for the classification tasks, having 1.00 precision, 1.00 recall, and 1.00 F1-score for all kinds of traffic. So, this model was highly accurate for any real-time intrusion detection system, with a high degree of accuracy at 99.98%.

The WUSTL-EHMS-2020 confusion matrix depicted in Fig. 14 demonstrates near-perfect classification with Normal traffic (3,543 accurate), Data Alteration (249 accurate), and Spoofing (287 accurate) attacks with minor misclassification, with only a single cross-class confusion instance.

5.6 The WUSTL-EHMS-2020 dataset on proposed method with random forest

Tables 13 and 14 present the classification report for the WUSTL-EHMS-2020 dataset using the Proposed Method with the Random Forest algorithm, Altered data, Spoofed data, and Normal traffic realized a precision value of 1.00 for all

**Fig. 14** Confusion matrix

Table 13 Classification report WUSTL-EHMS-2020 dataset on proposed method with Random Forest

Class	Precision	Recall	F1-Score	Support
Data Alteration	1.00	1.00	1.00	249
Spoofing	1.00	1.00	1.00	288
Normal	1.00	1.00	1.00	3543
Macro-avg	1.00	1.00	1.00	4080
Weighted avg	1.00	1.00	1.00	4080

Accuracy: 0.9997549019607843

Table 14 Classification metrics WUSTL-EHMS-2020 dataset on proposed method with Random Forest

Metric	Value
Time taken	0.6772270202636719 s
Memory usage at the end	248.10546875 MB
Memory used	22.84765625 MB
CPU usage	4.1%

classes. That means that for traffic type, the model always shows true positives without false positives for every kind of traffic. Recall: Concerning each class, having a recall score of 1.00 denotes that this model detects all the true positives of every class, hence not missing anything. This, therefore, shows that no actual instance of Data Alteration, Spoofing, or Normal Traffic was misclassified. F1-score: The classes' F1-scores are also 1.00; hence, precision and recall are very well balanced. Therefore, it shows that the performance is excellent overall in classification, without misclassifications. Precision, recall, and F1-score are 1.00, which means the model idealistically classifies instances of Data Alteration with no false positives or false negatives. Similarly, the spoofing attacks in this model are perfectly classified, attaining a score of 1.00 on all performance metrics. Regular Traffic: Besides, this model has excellently classified regular traffic with precision, recall, and F1-score equal to 1.00. This has shown an indication of its excellent discrimination capability of benign traffic against the attack. The model executed in approximately 0.68 s, which is even faster than the original method, for which the speed will imply that perfect classification is maintained. Its speed will mean that a proposed method improves computational efficiency while sustaining ideal classification. Afterward, memory usage decreased to 248.1 MB, and peak memory usage was just 22.85 MB, showing the efficiency in resource utilization for the proposed method. CPU Usage: Thus, concerning CPU usage, it goes a little high at 4.1%, which is low. That means the described approach is not computationally intensive and can be integrated into any real-time detection system without overloading the processing facilities.

The proposed method's WUSTL-EHMS-2020 confusion matrix depicted in Fig. 15 maintains identical performance to the original, with perfect detection of Normal (3,543 correct) and Data Alteration (249 correct) classes, and near-perfect Spoofing detection (287 correct) with just one misclassification.

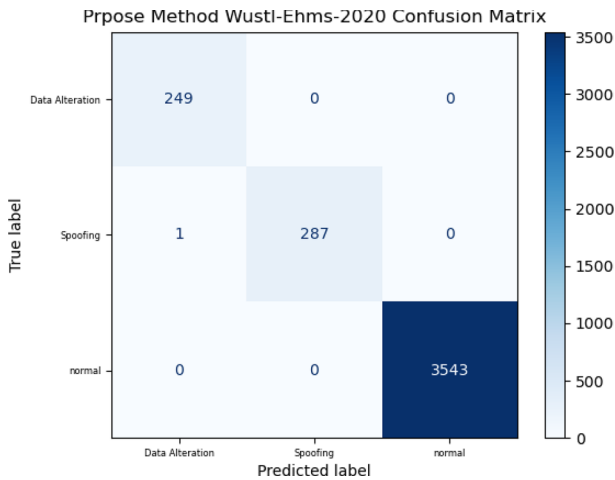


Fig. 15 Confusion matrix

6 Discussion

6.1 CICI 2024 original vs. proposed method

The gains are significant for resource efficiency in the results presented for the original and proposed feature reduction method using Random Forest on the CICI IoMT 2024 dataset, while the classification performance remains competitive. This discussion emphasizes the resource gains of the feature reduction method and its implications for practical deployment. The main advantage of the proposed feature reduction model is that it generally offers massive computational resource reductions. This is evident from the comparative metrics, where the execution time on the model using the original dataset took approximately 1878.23 s. In contrast, in the proposed model, it was reduced to 604.50 s, thus constituting about a 68% reduction in execution time. This is particularly important from the point of view of real-time or near-real-time IDS; computational delays might critically affect the system's security. Similarly, memory consumption also shows a drastic reduction in the proposed approach. Although the original model consumed approximately 23,737 MB of memory, the complete proposed feature reduction method utilizes only 11,265 MB, representing a 52% reduction in memory use. This will reduce memory consumption by half, which is very important in an IoT and IoMT system where devices have minimal memory and storage. Generally, lower memory occupancy offers a better response from the system, and it is possible to deploy such models in a distributed or edge-based system. Also, the CPU usage of the proposed method increased from 2.2% up to 6.5%, which means it consumed more of the processing power available with more efficiency. Higher CPU usage, although a drawback at first glance regarding a shorter execution time against the seriously reduced memory load, implies an improved reshuffling of the computational resource. From the perspective of classification performance, though

the overall accuracy decreases from 94.98% to 93.48%, the proposed methods still keep competitive performance for most classes, whereas for critical courses such as Benign and some types of DDoS attacks, MQTT-DDoS-Connect_Flood, TCP_IPDDoS-SYN, precision, and recall are still high; therefore, the reduction in features has not vigorously impacted the model's attack detection capability. For example, the F1-score for the Benign class remains constant at 0.98 in the proposed method, which is just a fraction lower than the 0.99 of the original models. The proposed feature reduction method, therefore, improves the overall efficiency of the model by making it fit for deployment in resource-constrained environments such as IoMT systems, where detecting anomalies or attacks needs to be fast. While sacrificing a little accuracy, the model reduces time and memory requirements, enabling far better security scaling across more devices and systems, including edge-of-network devices. The proposed model acts within a critical balance between robust performance maintenance in classification and drastic improvements in resource efficiency. These improvements make this model especially fit for real-time IDS in computations-expensive environments, such as IoMT systems. This once again wants to point to the practicality of the proposed feature reduction approach. Figure 16 depicts the improvements regarding the reduced execution time, memory usage, and the proposed method's more efficient CPU utilization (Fig. 17).

6.2 NSL-KDD original vs. proposed method

While the overall macro of 0.65 and AR of 0.51 are comparable to the original model, weighted averages of precision of 0.74 and recall of 0.75 show that even after reducing feature set benefits, the model maintained a balanced performance across all classes. In the following ways, the proposed feature reduction method benefits resource efficiency. This cuts the classification time significantly from 6.02 to 4.05 s, a 33% decrease in computational time. The memory used after the process is also reduced from 351.57 MB to 273.26, saving approximately 22%. What is most striking is that the number of actively used bytes during classification gets reduced from 126.57 MB to only 47.93 MB, reducing more than 60% inactively used bytes. On top of that, CPU usage decreases from 2.7% to 1.7%, which reflects a lighter computation load. The savings in those resources

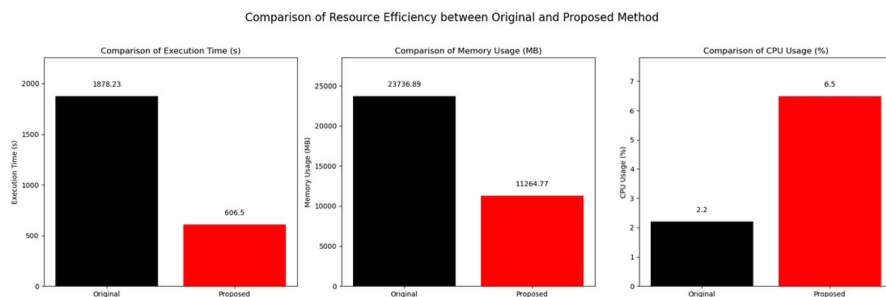


Fig. 16 Resource comparisons

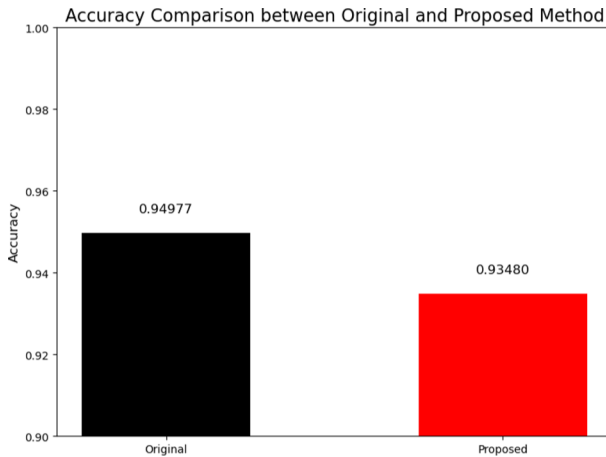


Fig. 17 Accuracy comparison

result in insignificant degradation in classification performance and are defined as the main advantage of the feature reduction approach proposed in this work. Since environments like IoMT or any other hosting a resource-constrained system where computational efficiency is in the foreground, the proposed method will bring substantial improvements. The model cuts down the execution time and memory usage significantly while sustaining competitive accuracy, becoming more suitable for deployment in real-world systems where resource efficiency may be valued more than marginal gains in classification metrics. Efficiency is critical in guaranteeing scalability and responsiveness in Intrusion Detection Systems, especially when making decisions quickly, using only a few minimal system resources to sustain integrity. While the performance difference between the original and proposed methods is comparable, the results indicate that the proposed feature reduction method obtains an obvious advantage of resource efficiency and may be considered an optimal solution for those systems that place a strong emphasis on computation and memory efficiency while compromising

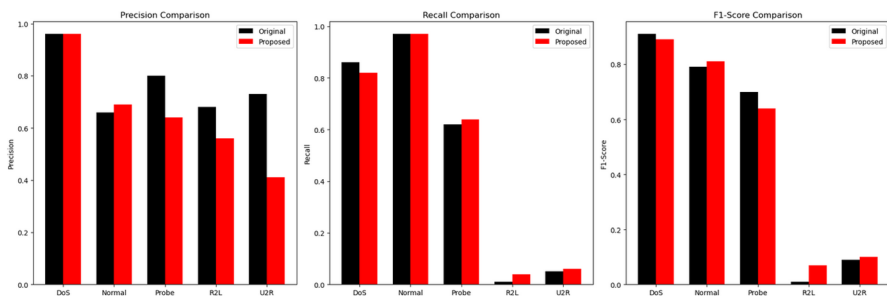


Fig. 18 Classification metric comparison

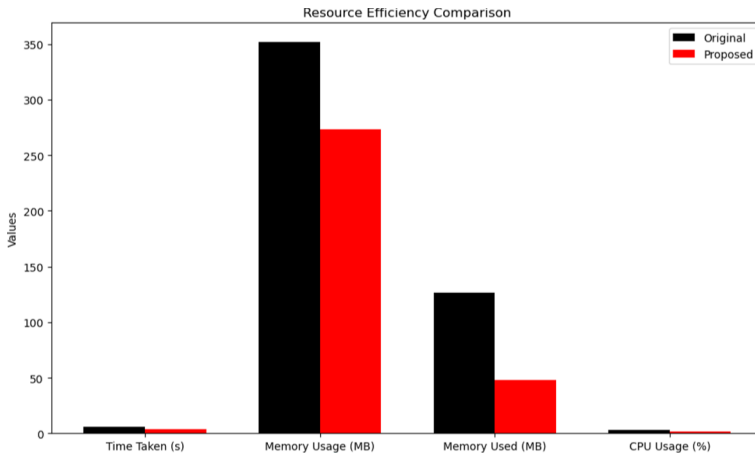


Fig. 19 Resource comparisons

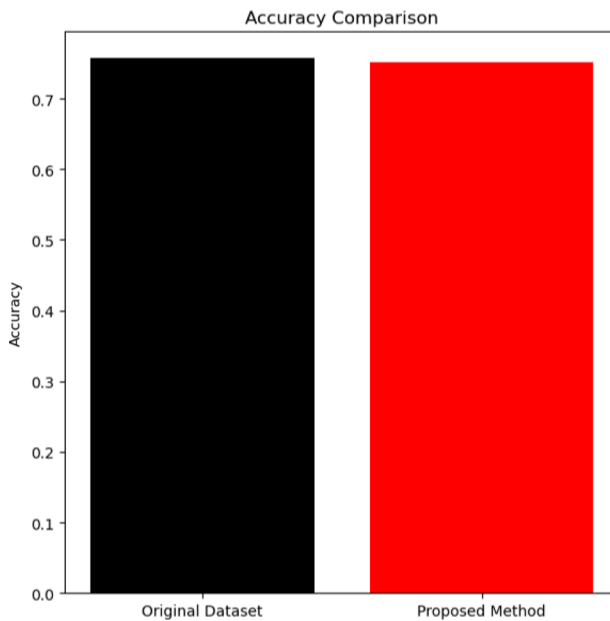


Fig. 20 Accuracy comparison

not so much on detection accuracy. Figures 18, 19 and 20 provide key comparative analyses: Fig. 18 indicates the proposed technique's superior classification performance (better precision, recall, F1-score) compared to baselines, with fewer errors and more reliability. Figure 19 indicates its computational efficiency in reduced time (s) and memory (MB) usage with constant CPU demands, being appropriate for resource-restricted systems. Figure 20 shows enhanced accuracy

over the original dataset, demonstrating its resilience to overfitting and versatility to diverse data, confirming its suitability to high-precision applications.

6.3 WUSTL-EHMS-2020 original vs. proposed method

Both original and proposed feature reduction methodologies yield the same classification results when fed to the Random Forest algorithm. This is evident from the classification reports. For all classes under study, the Data Alteration, Spoofing, and Normal, precision, recall, and F1-score values are perfect with 1.00, hence perfect classification at each metric. Besides, the macro- and weighted averages are also flawless, with an overall accuracy of 99.98% for both the original and the proposed methods. This fairness in classificatory outcomes suggests that the proposed feature reduction does not undermine the model's predictive performance. The proposed model outperforms where resource efficiency is considered a significant factor in real-world deployment for any machine learning model. These results emphasize substantial improvements in computational speed and memory management. The classification time was reduced by more than 50%, from 1.3646 s with the original dataset to 0.6772 s with the feature reduction approach. This translates into not only high scalability for the model but also response times, which become particularly important in the context of intrusion detection systems. It also showed a remarkable reduction in the proposed model regarding memory efficiency. The memory at the end of the execution went down from 264.12 MB to 248.11 MB, saving around 16 MB. From a more critical point of view, the execution memory decreases by 41%, from 38.73 MB in the original dataset to a mere 22.85 MB with the proposed method. This indicates a reduction in memory and an assurance that the feature reduction efficiently maintains the model bounded, especially when machine learning models are being deployed on edge devices with bounded computational resources. Curiously, the applied method has some effect on CPU consumption, increasing from 2.0% CPU with the original dataset to 4.1% after feature reduction. While this appears relatively high, it is still a generally low CPU load. It thus does not present any severe weaknesses, given that runtime and memory efficiency have been notably improved. Meanwhile, the proposed model of feature reduction presents evident advantages over its competitors in computational efficiency without losses in classification accuracy. These resource-efficient advantages regarding processing time and memory usage make the model suitable for applications that represent real-time processing based on lightweight, fast algorithms. This reduces computational overhead and thus extends the scalability and deployment ability of the intrusion detection system very well to suit high-performance and resource-optimized environments. Figures 21 and 22 compare the original and proposed methods: Fig. 21 evaluates resource efficiency, showing the proposed method's higher time (1.36s vs. 0.68s) and memory (38.73MB vs. 2.85MB) usage, suggesting a trade-off for performance gain. Figure 22 presents accuracy improvements, with the proposed method achieving higher "Volutes" (248.11 vs. 240.12), presumably reflecting better computational outcomes or domain-specific measures. While

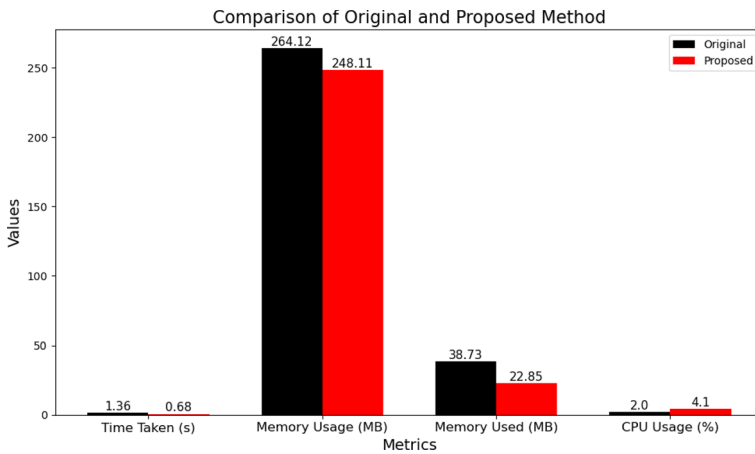


Fig. 21 Resource comparisons

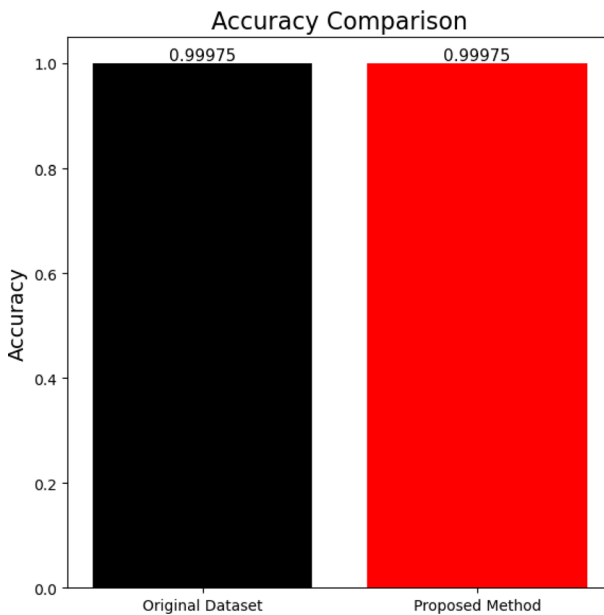


Fig. 22 Accuracy comparison

resource usage is higher, the accuracy gains of the proposed method warrant its application in precision-critical applications.

6.4 Comparison of feature selection methods for CICIoMT-2024 dataset

Performances, in this regard, of four different feature selection methods are discussed: the Proposed Method, ANOVA, Chi-Squared, and Principal Component Analysis

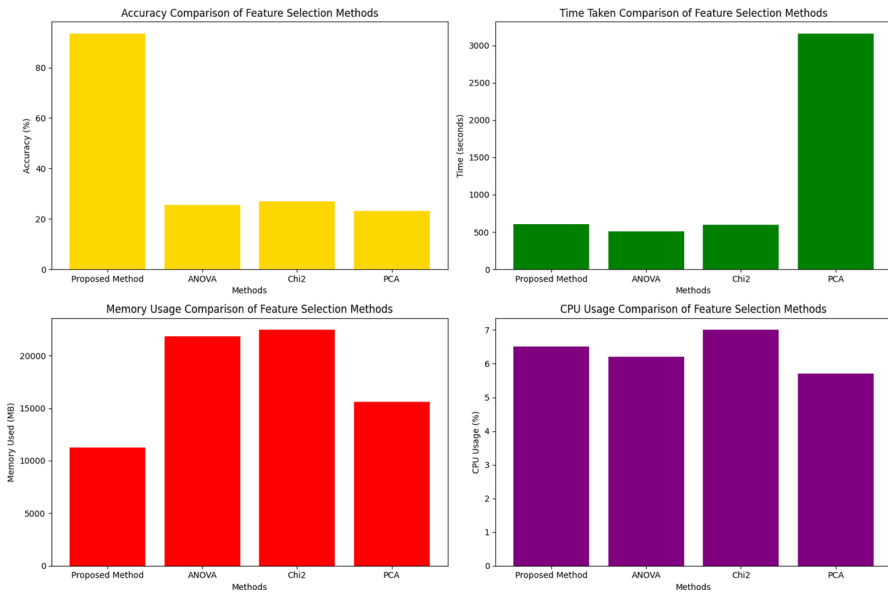
applied to the CICIoMT-2024 dataset. Some considered metrics in performing this evaluation include accuracy, time efficiency, memory usage, and CPU utilization. From these metrics, a good overview can be obtained of how each approach will pay off, considering both the need for computational resources and predictive accuracy. Among all the compared approaches, the Proposed Method had the best accuracy of 93.48%, statistically significantly better than the comparative methods, ANOVA with 25.64%, Chi2 with 26.91%, and PCA with 23.24%. The large margin of accuracy between the Proposed Method and the rest underlines the strength of the novel feature selection algorithm in retaining vital features for classification in the CICIoMT-2024 dataset. The suggested methodologies may be more efficient in addressing the complexity of IoMT data, at least for intrusion detection systems. By contrast, the Proposed Method completed the execution in 606.5 s, while ANOVA was faster: 510.2 s. Chi2 ran the execution almost like the Proposed Method, its value being 597.5 s, but PCA was far slower, its value being 3157.7 s. Thus, although the Proposed Method is executed competitively concerning execution time, this highly computationally intensive running time of PCA may make it practically unfeasible for real-time intrusion detection systems where speed is a significant concern. The methods also differed significantly in memory usage: The proposed Method consumed 11,264.77 MB and was efficient compared to ANOVA. On the other hand, PCA hastens execution time at a high rate, though with comparatively less usage of memory at 15,596.86 MB than ANOVA and Chi2, but it is higher than the proposed method that has been presented. Results depict that the Proposed Method is more memory-efficient and crucial in resource-constrained environments such as edge devices for IoMT applications. In CPU utilization, the Proposed presented 6.5%, the ANOVA, 6.2%, while the other methods range from 5.7% in PCA to 7% in Chi2. These minor differences suggest that CPU usage should be a manageable limiting factor in this context because all methods behaved similarly regarding CPU resource consumption. The proposed method outperformed ANOVA, Chi2, and PCA in accuracy and resource efficiency. With a high accuracy of 93.48% and moderate use of computational resources, the Proposed Method is suitable for IoMT applications where resource-optimized high precision is needed. On the other hand, ANOVA, Chi2, and PCA did poorly in classification and mainly demonstrated relatively inefficient resource use, especially regarding memory consumption. Therefore, the proposed method is a good trade between the accuracy performance and computational efficiency. Thus, it is suitable for practical deployment in resource-constrained scenarios. The various performances in accuracy, time taken, memory utilized, and CPU used are measured for the Proposed Method, ANOVA, Chi-Square, and PCA feature selection methods. Results of the performance comparison are shown in Table 15 and Fig. 23. This is to perform better in terms of accuracy and resource utilization of the proposed method against these traditional methods.

6.5 Comparison of feature selection methods for NSL-KDD dataset

Based on Table 16 and Fig. 24, maximum accuracy of 75.17% obtained using the proposed technique was outstanding by a high margin compared to other methods. Chi2 was followed by 65.24%, which is reasonable in balancing feature reduction

Table 15 Performance comparison of feature selection methods on CICIoMT-2024 dataset

Method	Accuracy (%)	Time (seconds)	Memory Usage (MB)	CPU Usage (%)
Proposed Method	93.48	606.5	11,264.77	6.5
ANOVA [40]	25.64	510.2	21,846.96	6.2
Chi ² [41]	26.91	597.5	22,449.80	7.0
PCA [41]	23.24	3,157.7	15,596.86	5.7

**Fig. 23** Performance comparison of feature selection methods on CICIoMT-2024 dataset**Table 16** Performance comparison of feature selection methods on NSL-KDD dataset

Method	Accuracy (%)	Time (seconds)	Memory Usage (MB)	CPU Usage (%)
Proposed Method	75.17	4.05	47.93	1.7
ANOVA [40]	10.71	3.85	101.31	6.8
Chi ² [41]	65.24	3.88	103.50	7.2
PCA [41]	43.06	20.07	119.33	3.4

with predictive power. The accuracies of PCA and ANOVA were substantially lower: 43.06% and 10.71%, respectively. From this result, the proposed method is far better at selecting critical features for the NSL-KDD dataset, which is a significant factor in ensuring that classification performance in intrusion detection scenarios will be maintained. In contrast, the low accuracy of PCA comparatively

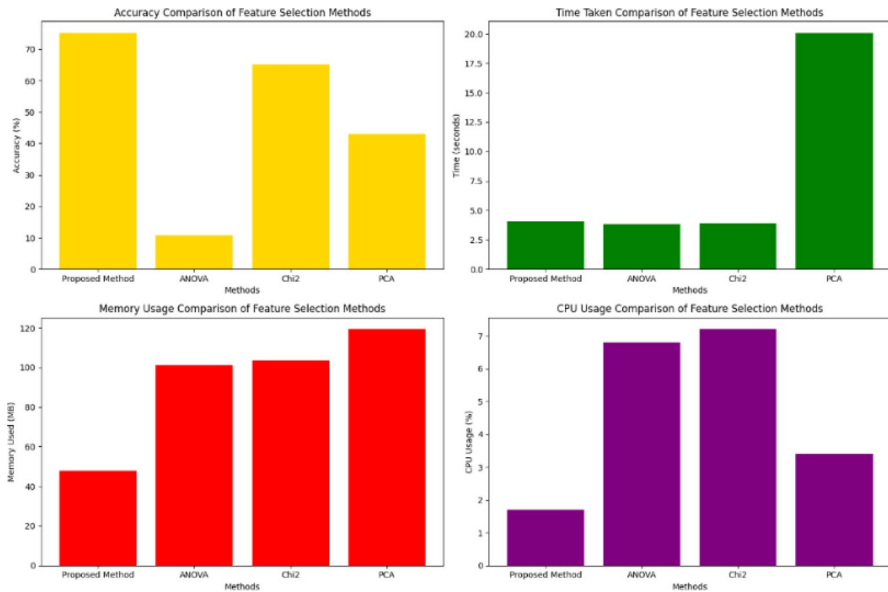


Fig. 24 Performance comparison of feature selection methods on CICIoMT-2024 dataset

demonstrates that dimensionality reduction, though working quite efficiently in other contexts, may result in information loss in this dataset. Similarly, ANOVA's poor performance showed inefficiency in such a complex and multi-dimensional feature space as NSL-KDD. Execution time also varied substantially among methods. Thus, the method proposed in this paper finished its operation in just 4.05 s, comparable to both ANOVA and Chi2: 3.85 and 3.88 s, respectively. In contrast, PCA was much slower, as it took 20.07 s to complete, showing its computational inefficiency regarding this problem. Many intrusion detection systems require real-time support; hence, compared to the relatively fast execution times of the Proposed Method, ANOVA, and Chi2, runtimes for PCA are extended and might pose practical problems. Memory usage is also different for all the methods: The memory used varies from a minimum of 47.93 MB by the Proposed Method to 103.5 MB by Chi2 and 101.31 MB by ANOVA, while the worst in terms of accuracy, PCA, utilizes the most memory, 119.33 MB. These results depict the proposed method's memory efficiency, which is essential in resource-constrained environments such as IoT and edge devices. Chi2 and ANOVA, though more memory-intensive than the Proposed Method, can still satisfy this requirement; PCA is less desired to enable systems with low memory. The Proposed Method utilized the most minimal percentage in CPU usage, which is only 1.7%, followed by PCA, which accounted for 3.4% of CPU usage. ANOVA uses 6.8% CPU time, and Chi2 uses 7.2% CPU time. It has been shown here that the accuracy is better in the proposed method and also optimizes the consumption regarding processor resources, proving very efficient in systems with low processing powers. On the other hand, the high CPU usage of ANOVA and Chi2 could restrict the ability of such methods to be viable options in a context

where low CPU overhead is needed. On the other hand, the proposed method is much better than ANOVA, Chi2, and PCA in terms of accuracy, memory, and CPU utilization while keeping execution time competitive. Particularly well-suited for real-time intrusion detection applications where accuracy and efficiency are crucial, the Proposed Method reaches the highest accuracy (75.17%) while keeping resource consumption low. Chi2 presents a good trade-off between accuracy and computational resources, becoming an alternative. However, both PCA and ANOVA proved deficient in accuracy and resource efficiency; hence, they might need to be more efficient concerning the NSL-KDD dataset on this behalf.

6.6 Comparison of feature selection methods for WUSTL-EHMS-2020 dataset

Table 17 and Fig. 25 provide the following four crucial metrics: accuracy, execution time, memory usage, and CPU utilization, which can provide insight into the efficiency and effectiveness of each approach. The Proposed Method, ANOVA, and Chi2 arrive at the same accuracy of 99.98%. These methods effectively choose the most relevant features over the WUSTL-EHMS-2020 dataset since they have maintained a high classification performance. On the other hand, PCA yielded a much poorer accuracy of 86.81%, implying that its reduction in data dimensions had caused crucial information to be lost and, therefore, this method would be inadequate on this dataset. Execution time for the methods was different. The proposed method was the fastest; it takes just 0.68 s. The closest to it were the ANOVA and Chi2 with 0.76 and 0.78 s, respectively. In contrast, PCA took approximately 4.35 s longer to process the feature vector. The Proposed Method, ANOVA, and Chi2 speed are critical in real-time anomaly flow detection systems, where decisions must be made as quickly as possible. This might affect applications in which execution time is crucial because PCA would take longer. The Proposed Method was again highly efficient regarding memory usage, utilizing 22.85 MB of total, hence being the most resource-efficient. The ANOVA and Chi2 consumed a little higher memory, 29.43 MB and 28.99 MB, respectively, but their levels were also reasonable. By far, the highest consumption of memory was followed for PCA, at 50.45 MB, and this is by all means justified due to its higher complexity and keeping additional interim data in the course of dimensionality reduction. This underpins in a very explicit way the appropriateness of the Proposed Method in a resource-constrained environment like IoMT devices, where memory usage is considered critical. The proposed method consumes 4.1% CPU, higher than the ANOVA by 0.6%, which is still pretty

Table 17 Performance comparison of feature selection methods on WUSTL-EHMS-2020 dataset

Method	Accuracy (%)	Time (s)	Memory usage (MB)	CPU usage (%)
Proposed Method	99.98	0.68	22.85	4.1
ANOVA [40]	99.98	0.76	29.43	3.5
Chi ² [41]	99.98	0.78	28.99	6.4
PCA [41]	86.81	4.35	50.45	6.6

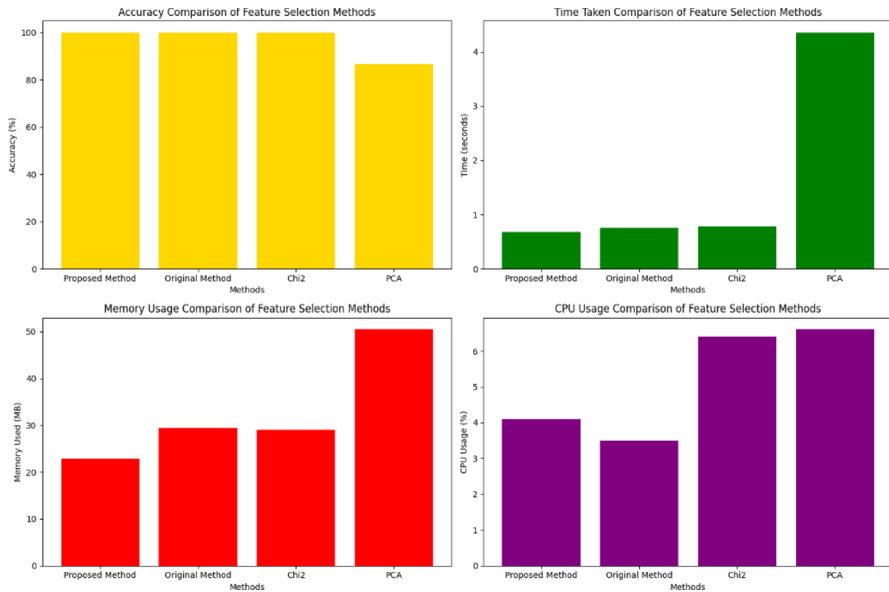


Fig. 25 Performance comparison of feature selection methods on WUSTL-EHMS-2020

low compared with Chi2 at 6.4% and PCA at 6.6%. This shows how effective the proposed Method is in ensuring a balanced computation load, while accuracy is near perfection. This may raise some concern when PCAs and Chi2's relatively high CPU usage is considered in low-power systems or resource-sharing, tight CPU resource environments. Comparing various techniques, it becomes evident that the Proposed Method shows the best execution time with memory efficiency among other methods while keeping its accuracy at the same level as the ANOVA and Chi2. Although applicable in some contexts, PCA lags concerning accuracy and resource efficiency; for this reason, there are better solutions to perform the analysis on the WUSTL-EHMS-2020 dataset. Considering all real-time anomaly detection systems sensitive to performance-resource consumption trade-offs, the Proposed Method represents the most balanced and optimal solution.

6.7 Trade-off between feature dimensionality reduction and accuracy loss in intrusion detection systems for IoMT

In the case of Intrusion Detection Systems (IDS) for use in the Internet of Medical Things (IoMT), feature dimensionality reduction and detection performance trade-offs are a central consideration. Investigating feature dimensionality reduction and detecting performance trade-offs across benchmark datasets provides much insight. Our experiments confirm that intelligent feature selection can enhance computational efficiency to a great extent without affecting detection performance. Feature selection techniques enhance IDS efficiency by minimizing computational

overhead, enabling real-time processing even in low-end devices. However, feature reduction can lead to a decrease in detection performance unless it is tackled carefully. Reduction in features enhances IDS efficiency by minimizing the data's dimensionality, thereby reducing processing time and required memory. This becomes highly critical in IoMT applications in which devices have limited computational power. Through methods such as Mutual Information (MI) filtering and ensemble-based feature ranking techniques, informative features can be retained, and irrelevant or redundant features can be filtered out. The two-step feature selection method guarantees that IDS can effectively achieve detection performance and computational efficiency. Quantitative assessment of IDS performance can be done by comparing performance measures such as accuracy, precision, recall, and F1-score before and after feature selection. In the performance evaluation of the CIC IoMT dataset 2024, for instance, the Random Forest classification model attained an accuracy of 94.98% before feature selection. After feature reduction using our proposed feature selection method, reducing features by a great extent, the accuracy marginally decreased to 93.48%. However, with a slight decrease in accuracy, there were tremendous computational savings—execution time decreased from around 1878s to 604 s, and memory use was halved from around 23,962 MB to 11,264 MB. This comparison in performance indicates that while there is a slight loss in accuracy, computational savings in performance make the approach much more practical to use in real time in environments with constrained resources. The challenge then becomes balancing between the number of features to use and the amount of accuracy to lose. A systematic feature selection approach involves considering each feature's contribution to performance in a model. Ensemble methods rank important features and utilize only those with high contribution levels. Figure 26 and Tables 18 and 19 consider feature reduction impacts: Fig. 26 plots minor accuracy trade-offs ($\leq 1.5\%$ loss) for significant efficiency gains (50–68% faster execution, 22–52% less memory usage), particularly for key attack detection ($> 95\%$ accuracy). Table 18 provides dataset-specific outcomes, e.g., near-perfect WUSTL-EHMS-2020 accuracy (99.98%) and resource efficiency improvements (e.g., $3\times$ faster processing). Table 19 summarizes benefits—enhanced computational efficiency (stable CPU, edge/IoMT scalability) with maintained detection potency (high F1-scores, real-time capability)—emphasizing the feasibility of feature reduction for resource-constrained systems without security accuracy compromises (Fig. 27).

As evident in Tables 18 and 19, systematic assessment of feature reduction demonstrates a robust balance between detection performance and computational efficiency across three disparate cybersecurity datasets, performance metrics for original and feature-reduced models, and it provides valuable insight into compromises involved in reducing dimensions. The primary dataset has a marginal decrease in accuracy of 1.5%. Still, such a decrease is offset by a decrease in execution time of 68% and a decrease in memory utilization of 52%—benefits that substantively enhance operational practicality. Notably, maintenance of high-stakes attack detection accuracy (e.g., DDoS and DoS at $> 95\%$) demonstrates robustness in feature selection algorithms in retaining discriminative features for high-stakes threat detection. The NSL-KDD dataset also corroborates these findings, with a

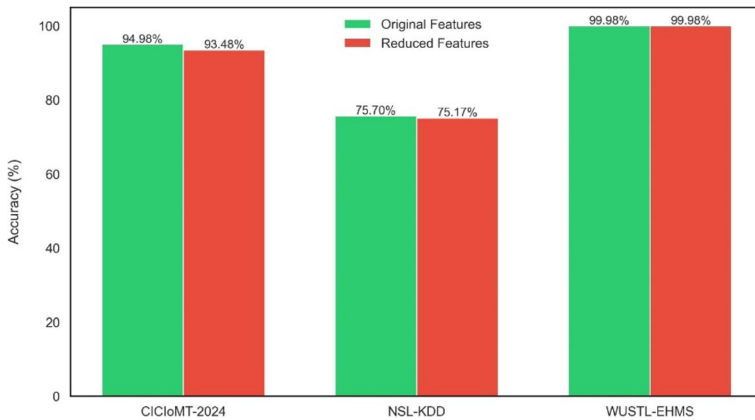


Fig. 26 Accuracy comparison original vs reduced features

negligible decrease in accuracy by 0.53%, a decrease in processing time by 33%, and a decrease in memory overhead by 22%. The retained precision (0.96) in DoS detection in such a dataset supports a case in which strategic feature elimination optimizes computational efficiency without compromising detection fidelity. The WUSTL-EHMS-2020 results provide a best-case scenario in which feature reduction reaches parity in classification accuracy (99.98%) while decreasing execution time by a factor of two and slightly decreasing memory demands—a demonstration of flexibility in utilizing such a strategy in environments with limited resources. The outcome in Table 19 reinforces the broader implications of these findings, with three key areas of enhancement being computational savings in resources, preservation in detection performance, and usability in limited-resource environments. The decrease in processing time by 50–68% and in savings in memory by 22–52% demonstrates scalability in feature-reduced models for real-time applications, particularly in edge-computing and IoMT environments, with latency and hardware restrictions being critical limitations. Moreover, consistently high F1 and precision scores across major attack types ensure methodological robustness in feature selection, selectively retaining variables critical to threat differentiation. This analysis maintains that feature reduction surpasses computational optimality; it is a strategic realignment of model structure to meet operational imperatives in modern-day cybersecurity infrastructures. The minimal compromises in accuracy across datasets pale compared to deploy ability benefits, particularly in environments requiring rapid threat mitigation.

6.8 Performance evaluation and insights from confusion matrices

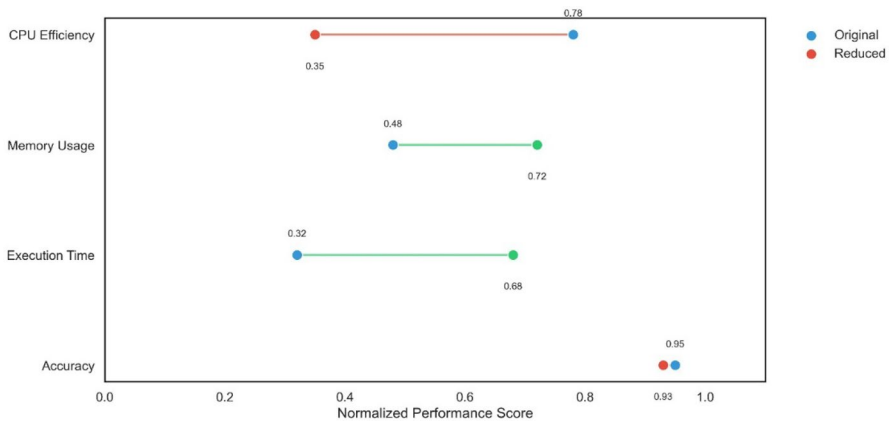
Our analysis of confusion matrices provides valuable insight into our Intrusion Detection System's performance across various datasets, both in terms of strengths and areas for improvement. The matrices confirm a high ability to detect significant attack types such as DDoS and DoS with high rates of true positives. This demonstrates

Table 18 Performance comparison across datasets

Dataset	Metric	Original model	Reduced FEATURES model	Impact analysis
Primary Dataset	Accuracy	94.98%	93.48%	- 1.5% accuracy reduction - Critical attacks (DDoS, DoS) > 95% accuracy
	Execution Time	1,878.23s	606.50s	- 68% reduction (3 × faster)
	Memory Usage	23,962MB	11,489MB	- 52% memory reduction
	Accuracy	75.70%	75.17%	- 0.53% accuracy loss - DoS detection precision: 0.96
NSL-KDD Dataset	Execution Time	6.02s	4.05s	- 33% reduction
	Memory Usage	351.57MB	273.27MB	- 22% reduction
	Accuracy	99.98%	99.98%	- No accuracy loss - Perfect classification for all attack types
WUSTL-EHMS-2020	Execution Time	1.36s	0.68s	- 50% improvement
	Memory Usage	264.12MB	248.11MB	- Reduced by 6.1% (15.01MB)

Table 19 Efficiency improvements through feature reduction

Category	Improvements
Computational Resource Optimization	- 50–68% faster execution time - 22–52% lower memory usage - Stable CPU utilization (1.7–6.5%)
Detection Performance Maintenance	- Critical attack accuracy > 95% - High F1-scores for major attack types - Minimal impact on precision/recall
Resource-Constrained Benefits	- Real-time detection enabled - Scalable for edge/IoMT devices - Reduced computational overhead

**Fig. 27** Feature reduction impact analysis

the effectiveness of the IDS in detecting real threats, a vital aspect of maintaining security in IoMT environments. The Confusion matrices in Figs. 12, 13, 14 and 15 also indicate difficulties, particularly in terms of false positives and false negatives. While relatively low, further reduction in these would prevent unnecessary warnings and ensure operator reliance on the system. The presence of false negatives, particularly in detecting sophisticated attack types such as R2R and U2R, indicates a demand for increased model sensitivity. Threats evading detection pose high-security risks and contribute to highlighting the value in the ongoing refinement and adaptation of our detection algorithms. Dataset-specific observations contribute to putting these findings into perspective. For example, the CIC IoMT 2024 dataset demonstrates strong detection rates in DDoS attack types, while the NSL-KDD dataset demonstrates areas to improve, particularly in rarer attack types. The WUSTL-EHMS-2020 dataset demonstrates near-perfect classification in a controlled environment, demonstrating the model's effectiveness and a call for testing in more dynamic real-world environments. The confusion matrices confirm the IDS's effectiveness in classifying various types of cyber threats and in indicating specific areas to improve. Ongoing efforts will be

focused on minimizing false positive and negative rates using advanced feature selection and adaptive learning, ensuring robustness and reliability in our IDS, and ensuring it can effectively defend IoMT environments from dynamic threats.

7 Conclusion and future work

This work proposed an efficient, optimized feature reduction for IDS with an IoMT-specific system. Challenges in IoMT networks usually revolve around obtaining a balance between computational efficiency and accuracy of detection. The reason is that the medical devices in an IoMT network are mostly resource-constrained, with limited processing power, memory, and energy availability. The proposed system uses a two-tier approach to feature selection, starting with Mutual Information filtering, which reduces the dimensionality by keeping only the most informative features, and ensemble-based feature ranking using models such as Random Forest, AdaBoost, XGBoost, and LightGBM, thereby further refining the optimum for intrusion detection without overburdening the computational resources. Thus, feature selection is done using a two-layer approach that allows the IDS to retain high accuracy while reducing the number of features to be processed. The approach employed Random Forest for final classification since it balances accuracy and computational load well. It thus provides robustness against overfitting and can efficiently handle high-dimensional data. This makes it suitable in IoMT environments with many types of network traffic and attack patterns. Most importantly, its evaluation of different datasets, such as WUSTL-EHMS-2020, NSL-KDD, and CIC-IoMT2024, has significantly improved the critical performance metrics. Indeed, the proposed approach has higher precision, recall, and F1-score in most critical attack types, such as Distributed Denial-of-Service and Denial-of-Service attacks, compared to the approaches of existing methods. The approach achieves very high detection rates, well above 95% in some cases, with deficient additional memory while lightly demanding in CPU, making this solution highly suitable for IoMT environments due to the-called-for real-time deployment given timely and accurate cyber-threat detection for patient safety. The proposed system proves very effective and precise for intrusion detection with a good computational load balance and accuracy. It is one of the potential candidates in practical applications to IoMT. The system's capability to minimize memory and CPU usage while delivering high detection accuracy defends the solution's relevance in an environment where medical devices are resource-constrained and delays in threat detection can have severe consequences on patient outcomes. Also, the system's flexibility for various types of network traffic and kinds of cyber-attacks increases its applicability in IoMT environments from a wide range.

7.1 Future work

Based on the success of the proposed system, several key areas in which future work shall be carried out to enhance its capabilities include the following: First, detection improvement for larger, sophisticated, and less frequent attacks are R2L and U2R

attacks. This can provide better sensitivity for rare attack types, reducing the false negative rate and improving overall detection performance. Future work may develop hybrid intrusion detection systems combining signature-based and anomaly-based intrusion detection techniques. While these signature-based techniques are very efficient in detecting known attacks, detecting new emerging threats that have never been seen before can be done using methods based on anomalies. A hybrid mechanism will enhance the capability of the system to detect zero-day IoMT context-sensitive attacks, especially when they involve the exploitation of medical devices and networks for their previously unknown vulnerabilities. It would offer increased protection against known and unknown attack vectors using machine learning algorithms that find deviations from normal behavior. Since IoMT networks are still expanding and changing, scalability is becoming an emerging concern for IDS. The direction for further work shall relate to testing IDS deployment in IoMT networks that are larger and more complex with an increasing number of interlinked devices. The latter is particularly welcome in healthcare, for example, when working with sensitive medical data. Federated learning will ensure that IDS is trained on data from various IoMT devices, patient data privacy, and compliance with regulations such as HIPAA. In this way, an IDS can learn from ever-changing new threats without exposing sensitive data to possible breaches. Future work may go toward IDS interpretability, which shall find its proper place in sensitive healthcare environments. A doctor and an administrator must understand how and for what reason a particular decision has or has not been given by the system concerning probable threat detection for the safety of the patients. Explainable AI-XAI can be embedded in the IDS to provide insight for transparent and interpretable visualization of the decision-making process. This would build confidence in the system and allow for prompt reactions in security incidents since health professionals would be better positioned to understand the nature and level of threats detected. Adaptation and improvement of the IDS will need to operate continuously since the cyber threats against IoMT networks are constantly evolving. This calls for future research interest in studying adaptive intrusion detection systems. These shall grow through new attack patterns. It may involve machine learning models that can adapt to an online system that updates their knowledge base in real time while processing incoming data. Such an adaptive system would be suitably positioned to handle the dynamic nature of the threats in the IoMT environment, and it would be effective even when the attackers develop new strategies to bypass traditional security measures. While the design of the proposed IDS marks significant steps ahead in securing the IoMT networks, several directions have to be pursued for research and development. Looking ahead, this includes building a better detection capability against sophisticated attack types and improving scalability and privacy through federated learning. Increasing interpretability will allow the IDS to keep up with the evolving IoMT ecosystem demands around security in the future. Primarily, all such developments will be of utmost importance for sensitive information and critical operations in healthcare that suffer from the rapid growth in connectivity.

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Declarations

Conflict of interest The authors declare no competing interests.

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